

The Cell as a Unit of Health and Disease

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Pathology literally translates as the *study of suffering* (Greek *pathos* = suffering, *logos* = study); more prosaically, and as applied to modern medicine, the *study of disease*. Virchow was prescient in asserting that disease originates at the cellular level, but we now appreciate that cellular pathologies arise from perturbations in molecules (genes, proteins, and metabolites) that influence cell survival and behaviors. Thus, the foundation of modern pathology is understanding the *cellular and molecular* aberrations that give rise to diseases. It is illuminating to consider these abnormalities in the context of normal cellular structure and function, which is the subject of this introductory chapter.

It is unrealistic (and even undesirable) to condense the vast and fascinating field of cell biology into a single chapter. Consequently, rather than attempting a comprehensive review, this chapter surveys basic principles and highlights recent advances relevant to the mechanisms of disease that are emphasized throughout the rest of the book.

THE GENOME

The sequencing of the human genome at the beginning of the twenty-first century represented a landmark achievement of biomedical science. Since then, the rapidly declining cost of sequencing, the burgeoning computational capacity to mine the ensuing data, and the expanding toolkits to analyze functional outputs (genomics, proteomics, and metabolomics) promise to revolutionize our understanding of health and disease. The emerging information has also revealed a breathtaking level of complexity far beyond the linear sequence of the genome. The potential of these powerful innovations to explain disease

pathogenesis and drive therapeutic discovery excites and inspires scientists and the lay public alike. The genome contains both coding and noncoding DNA. The significance of the latter is only beginning to be realized, as discussed below.

Noncoding DNA

The human genome contains some 3.2 billion DNA base pairs. Yet, within the genome there are only about 20,000 to 25,000 protein-encoding genes, constituting just 1.5% of the genome. Although 20,000 underestimates the actual number of encoded proteins (many genes produce multiple RNA transcripts that translate to different protein isoforms), it is nevertheless startling to realize that worms, which are composed of fewer than 1000 cells and have 30-fold smaller genomes, also have about 20,000 protein-encoding genes. Many of these proteins are recognizable homologs of molecules expressed in humans. What then separates humans from worms?

The answer is not completely known, but evidence suggests that much of the difference lies in the 98.5% of the human genome that does not encode proteins. The function of such long stretches of DNA (so-called genome “dark matter”) was mysterious for many years. However, over 85% of the human genome is ultimately transcribed, and nearly 80% is devoted to regulation of gene expression. It follows that while proteins provide the building blocks and machinery required for assembling cells, tissues, and organisms, it is the noncoding regions of the genome that provide the critical “architectural planning.” Practically stated, the difference between worms and humans apparently lies more in their genomic “blueprints” than in their construction materials.

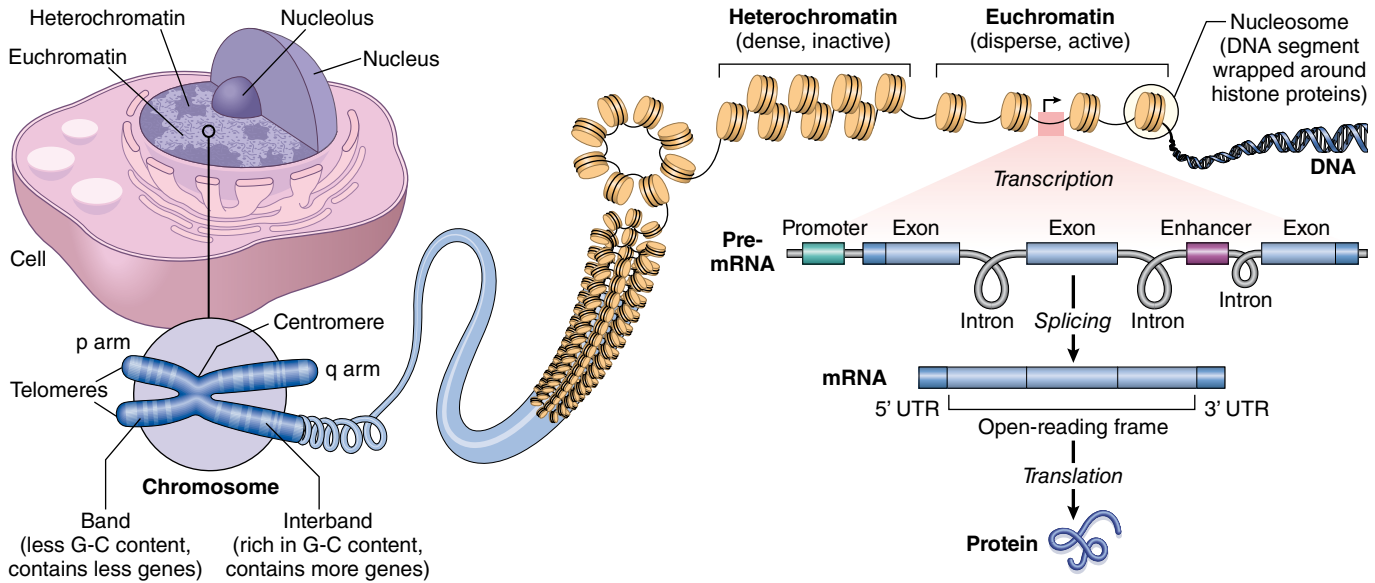


Fig. 1.1 Organization of nuclear DNA. At the light microscopic level, nuclear genetic material is organized into dispersed, transcriptionally active euchromatin and densely packed, transcriptionally inactive heterochromatin. Chromosomes (as shown) can be visualized only during mitosis, where they are organized into paired chromatids connected at centromeres. The centromeres act as the locus for the formation of a kinetochore protein complex that regulates chromosome segregation at metaphase. The *telomeres* are repetitive nucleotide sequences that cap the termini of chromatids. The chromatids are organized into short “p” (“petite”) and long “q” (the next letter in the alphabet) arms. Individual chromatin fibers are comprised of a string of nucleosomes—DNA wound around octameric histone cores—with the nucleosomes connected via DNA linkers. Promoters are noncoding regions of DNA that initiate gene transcription; they are on the same strand and upstream of their associated gene. Enhancers can modulate gene expression over distances of 100 kb or more by looping back onto promoters and recruiting additional factors that drive the expression of pre-messenger RNA (*pre-mRNA*) species. Intronic sequences are spliced out of the pre-mRNA to produce the final mature mRNA that contains an open reading frame that is translated into protein and 3′ and 5′ untranslated regions (*UTRs*) that often contain regulatory elements. G-C, Guanine-cytosine.

There are five major classes of functional non-protein-coding sequences in the human genome (Fig. 1.1):

- *Promoter and enhancer* regions that provide binding sites for transcription factors.
- Binding sites for factors that organize and maintain higher order chromatin structures.
- *Noncoding regulatory RNAs*. Over 60% of the genome is transcribed into RNAs that are not translated but regulate gene expression through a variety of mechanisms. The two best-studied varieties—micro-RNAs (miRNAs) and long noncoding RNAs (lncRNAs)—are described later.
- Special structural regions of DNA, in particular, *telomeres* (chromosome ends) and *centromeres* (chromosome “tethers”). A major component of centromeres is so-called *satellite DNA*, consisting of large arrays—up to megabases in length—of repeating sequences (from 5 bp up to 5 kb). Although classically associated with spindle apparatus attachment, satellite DNA is also important in maintaining the dense, tightly packed organization of heterochromatin (discussed later).
- *Mobile genetic elements* (e.g., *transposons*) make up more than one-third of the human genome. These “jumping genes” can move around the genome during evolution, resulting in variable copy number and positioning even among closely related species (e.g., humans and other primates). Although implicated in gene regulation and chromatin organization, the function of mobile genetic elements is not well established.

Many genetic variations (polymorphisms) associated with diseases are located in non-protein-coding regions of the genome. Thus, variation

in gene regulation may be more important in disease causation than structural changes in specific proteins. Another surprise that emerged from genome sequencing is that any two humans are typically more than 99.5% DNA-identical (and are 99% sequence-identical with chimpanzees)! Thus, individual variation, including differential susceptibility to diseases and environmental stimuli, stems from differences in less than 0.5% of our DNA (representing about 15 million bp).

The two most common forms of DNA variation in the human genome are single nucleotide polymorphisms (SNPs) and copy number variations (CNVs).

- SNPs are variants at single nucleotide positions and are almost always biallelic (only two choices exist at a given site within the population, such as A or T). Over 6 million human SNPs have been identified, with many showing wide variation in frequency in different populations.
 - SNPs occur throughout the genome—within exons, introns, intergenic regions, and coding regions.
 - Roughly 1% of SNPs occur in coding regions, which is about what is expected by chance, since coding regions comprise about 1.5% of the genome.
 - SNPs located in noncoding regions can occur within genomic regulatory elements, thereby altering gene expression; in such instances, SNPs influence disease susceptibility directly.
 - Some SNPs, termed “neutral” variants, are thought to have no effect on gene function or individual phenotype.
 - Even “neutral” SNPs may be useful markers if they happen to be coinherited with a disease-associated polymorphism

because of physical proximity, a relationship referred to as *linkage disequilibrium*.

- The effect of most SNPs on disease susceptibility is weak, and it remains to be seen if identification of such variants, alone or in combination, can be used to develop effective strategies to identify those at risk and, ultimately, prevent disease.
- CNVs are a form of genetic variation consisting of different numbers of large contiguous stretches of DNA; these can range from 1000 base pairs to millions of base pairs. CNVs can be biallelic and simply duplicated or, alternatively, deleted in some individuals. At other sites there are complex rearrangements of genomic material, with multiple variants in the human population.
- CNVs are responsible for between 5 million and 24 million base pairs of sequence difference between any two individuals.
- Approximately 50% of CNVs involve gene-coding sequences; thus CNVs may underlie a large portion of human phenotypic diversity.

It is important to note that alterations in DNA sequence cannot by themselves explain the diversity of phenotypes in human populations; moreover, classic genetic inheritance cannot explain differing phenotypes in monozygotic twins. The answers to these conundrums probably lie in *epigenetics*—heritable changes in gene

expression that are not caused by variations in DNA sequence (see the next section).

Histone Organization

Even though virtually all cells in the body have the same genetic composition, differentiated cells have distinct structures and functions that arise because of lineage-specific gene expression programs. Such cell type-specific differences in transcription and translation depend on *epigenetic factors* (literally, factors that are “above genetics”) that can be conceptualized as shown in Fig. 1.2:

- *Histones and histone-modifying factors.* Nucleosomes consist of DNA segments 147 bp long that are wrapped around a central core structure of highly conserved low-molecular-weight proteins called *histones*. The resulting DNA-histone complex resembles a series of beads joined by short DNA linkers. The naked DNA of a single human cell is about 1.8 m long. By winding around histones, like spools of thread, the entire genome can be packed into a nucleus as small as 7 to 8 μm in diameter. In most cases, this structured DNA, termed *chromatin*, is not wound uniformly. Thus, at the light microscopic level, nuclear chromatin is recognizable as cytochemically dense and transcriptionally inactive heterochromatin and as disperse transcriptionally active euchromatin (see

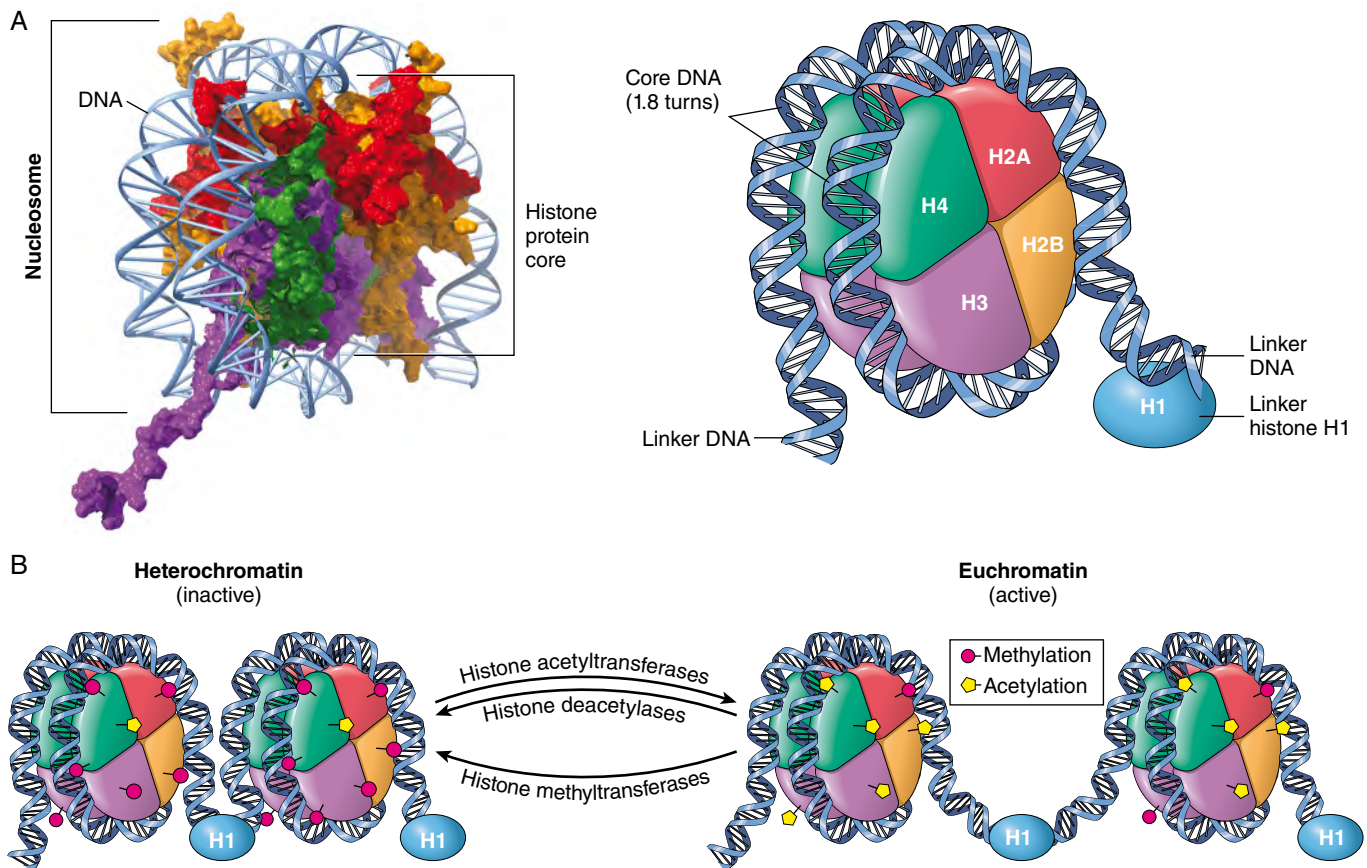


Fig. 1.2 Histone organization. (A) Nucleosome consists of DNA segments wrapped around a histone octamer protein (two each of histone subunits H2A, H2B, H3, and H4). Another histone, histone H1, sits on 20- to 80-nucleotide stretches of linker DNA between nucleosomes. Histone subunits are positively charged, thus allowing compaction of negatively charged DNA. (B) The relative state of DNA unwinding (and thus access for transcription factors) is regulated by histone modifications, including acetylation, methylation, and/or phosphorylation. These “marks” are dynamically written and erased; certain marks such as histone acetylation “open up” the chromatin structure, whereas others such as methylation of particular histone residues condense DNA to silence genes.

Fig. 1.1). In general, only the regions that are “unwound” are available for transcription. Chromatin structure can therefore regulate transcription independent of DNA regulatory elements (e.g., promoters and enhancers) and, due to variation between cell types, define cellular identity.

Histones are not static, but rather are highly dynamic structures regulated by a host of nuclear proteins. Thus, chromatin-remodeling complexes can reposition nucleosomes on DNA, exposing (or obscuring) gene regulatory elements such as promoters. “Chromatin writer” complexes, on the other hand, carry out over 70 different histone modifications, generically denoted as “marks,” which include methylation, acetylation, or phosphorylation of specific amino acids within histones.

Actively transcribed genes in euchromatin are associated with histone marks that make the DNA accessible to RNA polymerases. In contrast, inactive genes have histone marks that enable DNA compaction into heterochromatin. Histone marks are reversible through the activity of “chromatin erasers.” Still other proteins function as “chromatin readers,” binding histones that bear particular marks and thereby regulating gene expression.

- **Histone methylation.** Both lysines and arginines can be methylated by specific writer enzymes; methylation of histone lysine residues can lead to transcriptional activation or repression, depending on which histone residue is marked.
- **Histone acetylation.** Lysine residues are acetylated by histone acetyltransferases (HATs), whose modifications tend to open the chromatin and increase transcription. In turn, these changes can be reversed by histone deacetylases (HDACs), leading to chromatin condensation.
- **Histone phosphorylation.** Serine residues can be modified by phosphorylation; depending on the specific residue, the associated DNA may be opened for transcription or condensed and inactive.
- **DNA methylation.** High levels of DNA methylation in gene regulatory elements typically result in transcriptional silencing. Like histone modifications, DNA methylation is tightly regulated by methyltransferases, demethylating enzymes, and methylated-DNA-binding proteins.
- **Chromatin organizing factors.** Much less is known about these proteins, which are believed to bind to noncoding regions and control long-range looping of DNA, thus regulating the spatial relationships between enhancers and promoters that control gene expression.

Deciphering the mechanisms that allow epigenetic factors to control genomic organization and gene expression in a cell type-specific fashion is an extraordinarily complex proposition. Despite the intricacies, there is already ample evidence that dysregulation of the “epigenome” has a central role in malignancy (see Chapter 7), and emerging data indicate that many other diseases are associated with inherited or acquired epigenetic alterations. Unlike genetic changes, many epigenetic alterations (e.g., histone acetylation and DNA methylation) are reversible and amenable to therapeutic intervention, and several drugs that target epigenetic regulators (e.g., DNA methylation inhibitors) are used currently in the treatment of cancer.

MicroRNA and Long Noncoding RNA

Genes can also be regulated by noncoding RNAs. These genomic sequences are transcribed but not translated. Although many distinct families of noncoding RNAs exist, we will discuss only two examples here: small RNA molecules called *microRNAs* (*miRNAs*) and *long noncoding RNAs* (*lncRNAs*) (>200 nucleotides in length).

MicroRNA

miRNAs do not encode proteins; they modulate translation of target messenger RNAs (mRNAs). Posttranscriptional silencing of gene

expression by miRNA is a fundamental and well-conserved mechanism of gene regulation present in all eukaryotes (plants, animals, and fungi). Even bacteria have a primitive version of the same machinery that they use to protect themselves against foreign DNA (e.g., phage and virus DNA). The profound influence of miRNAs on protein expression allows these relatively short RNAs (22 nucleotides on average) to be critical regulators of developmental pathways as well as pathologic conditions (e.g., cancer).

The human genome encodes almost 6000 miRNA genes. Individual miRNAs can regulate multiple protein-coding genes, allowing each miRNA to coregulate entire programs of gene expression. Transcription of miRNA genes produces a primary transcript (pri-miRNA) that is processed into progressively smaller segments, including trimming by the enzyme Dicer. This generates mature single-stranded miRNAs of 21 to 30 nucleotides that associate with a multiprotein aggregate called *RNA-induced silencing complex* (*RISC*) (Fig. 1.3). Subsequent base pairing between the miRNA strand and its target mRNA directs the RISC to either induce mRNA cleavage or repress its translation. In this way the target mRNA is *posttranscriptionally silenced*.

Small interfering RNAs (siRNAs) are short RNA sequences that can be introduced experimentally into cells where they serve as substrates for Dicer and interact with RISC, thereby reproducing endogenous miRNAs function. Synthetic siRNAs that target specific mRNA species are powerful laboratory tools to study gene function (so-called knockdown technology) and several have now been approved for clinical use in diseases caused by excessive or inappropriate gene function, such as certain forms of amyloidosis (Chapter 6).

Long Noncoding RNA

Recent studies have further identified an untapped universe of lncRNAs, which by some calculations may exceed coding mRNAs by 10-fold to 20-fold. lncRNAs modulate gene expression by several mechanisms (Fig. 1.4). As one example, lncRNAs can bind to chromatin and restrict RNA polymerase from accessing coding genes within that region. The best-known example is XIST, which is transcribed from the X chromosome and plays an essential role in the physiologic X chromosome inactivation that occurs in females. XIST itself escapes X inactivation but forms a repressive “cloak” on the X chromosome from which it is transcribed, resulting in gene silencing. Conversely, many enhancers are sites of the synthesis of lncRNAs that increase transcription from gene promoters via a variety of mechanisms (see Fig. 1.4).

Gene Editing

An exciting new development that allows high-fidelity genome editing is ushering in the next era of the molecular revolution. This advance comes from a wholly unexpected source: the discovery of *clustered regularly interspaced short palindromic repeats* (*CRISPRs*) and CRISPR-associated genes (*Cas*), such as the Cas9 nuclease. These are linked genetic elements that endow prokaryotes with a form of acquired immunity to phages and plasmids. Bacteria use the system to sample the DNA of infecting agents and integrate portions into their genomes as CRISPRs. These CRISPR segments are subsequently transcribed and processed into guide RNA sequences that bind and direct the Cas9 nuclease to specific sites (e.g., a phage sequence) so that it can be cleaved to disable the infecting agent.

Gene editing repurposes this process by using artificial 20-base guide RNAs (gRNAs) that bind Cas9 and are complementary to a targeted DNA sequence (Fig. 1.5). Cas9 then induces double-stranded DNA breaks at the site of gRNA binding. Repair of the highly specific cleavages can lead to random disruptive mutations (through nonhomologous end joining) or can introduce new genetic material with precision (by homologous recombination). Both the guide sequences

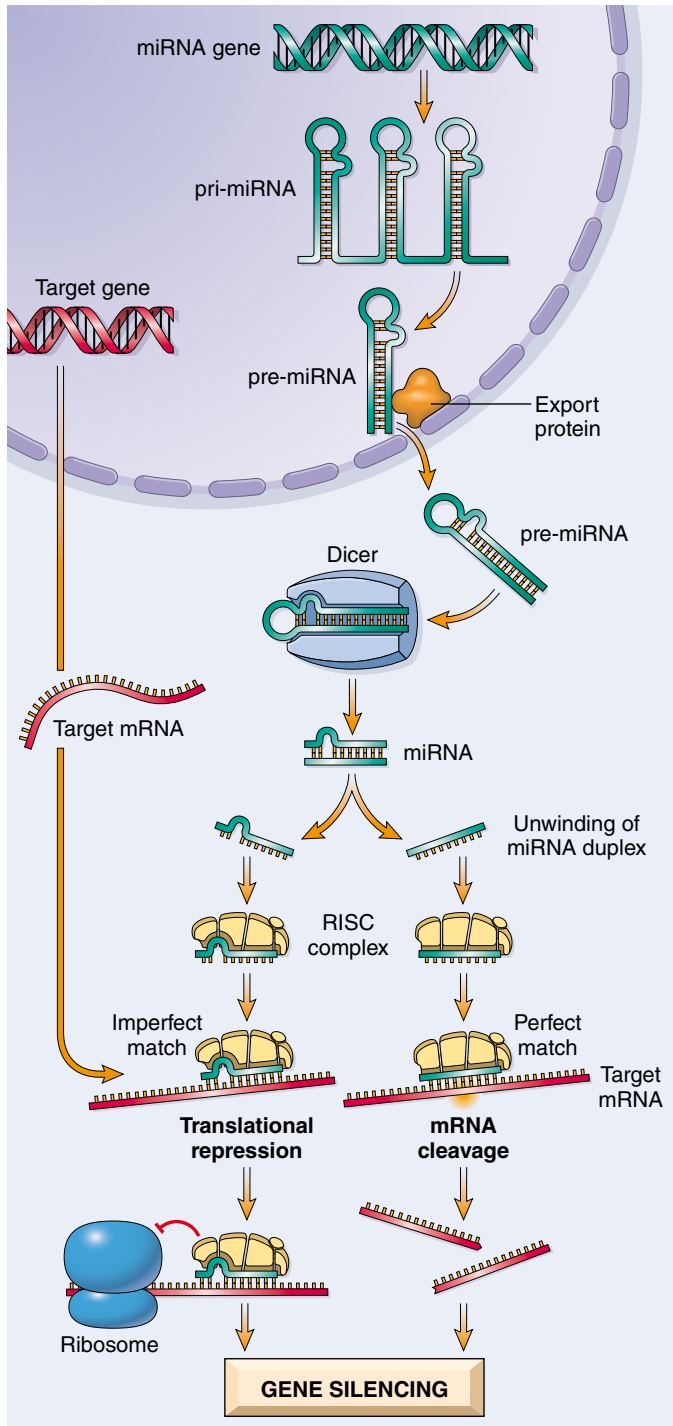


Fig. 1.3 Generation of microRNAs (*miRNAs*) and their mode of action in regulating gene function. Transcription of a miRNA produces a primary miRNA (*pri-miRNA*), which is processed within the nucleus to form *pre-miRNA* composed of a single RNA strand with secondary hairpin loop structures and stretches of double-stranded RNA. After export out of the nucleus via specific transporter proteins, *pre-miRNA* is trimmed by the cytoplasmic enzyme Dicer to generate mature double-stranded miRNAs of 21 to 30 nucleotides. The miRNA subsequently unwinds, and the single strands are incorporated into multiprotein RNA-induced silencing complexes (*RISC*). Base pairing between single-stranded miRNA and the targeted messenger RNA (*mRNA*) directs *RISC* to either cleave or repress translation of the mRNA, resulting in posttranscriptional silencing.

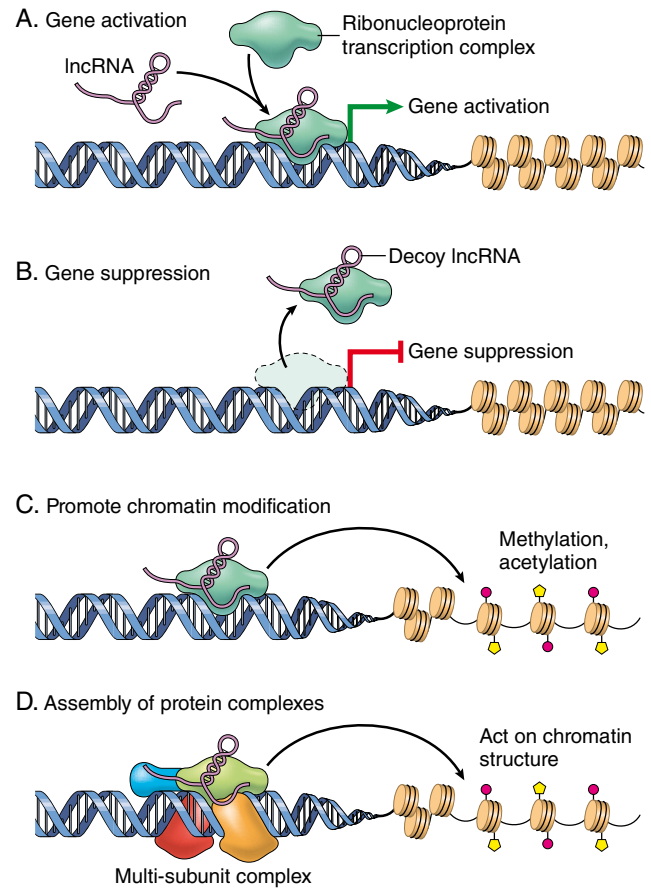


Fig. 1.4 Roles of long noncoding RNAs (*lncRNAs*). (A) lncRNAs can facilitate transcription factor binding, thus promoting gene activation. (B) Conversely, lncRNAs can preemptively bind transcription factors to inhibit transcription. (C) lncRNA binding can direct histone and DNA modification by acetylases (yellow dots) or methylases (pink dots) (or deacetylases and demethylases). (D) lncRNAs can act as scaffolds to stabilize secondary or tertiary structures and multi-subunit complexes that influence chromatin architecture or gene activity. (Modified from Wang KC, Chang HY. Molecular mechanisms of long noncoding RNAs. *Mol Cell*. 2011;43(6):904–914.)

and the Cas enzyme, either as a coding DNA (cDNA) or a protein, can be easily introduced into cells. The potential application to genetic engineering, due to the impressive specificity of the Cas9 system (up to 10,000-fold better than other previous editing systems), has led to great excitement. Applications include inserting specific mutations in cells and tissues to model cancers and other diseases, rapid generation of transgenic animal models from edited embryonic stem cells, and, recently, treatment of human inherited diseases such as sickle cell anemia (Chapter 14). CRISPR also makes it possible in principle to eliminate less “desirable” traits. Predictably, the technology has inspired a vigorous debate regarding the ethics of its use.

CELLULAR HOUSEKEEPING

Normal cell function and intracellular homeostasis depend on a variety of fundamental housekeeping functions that all differentiated cells must perform to maintain viability and normal activity. These include nutrient acquisition, metabolism, communication, movement, renewal of senescent molecules, molecular catabolism, energy generation, and protection from environmental stressors.

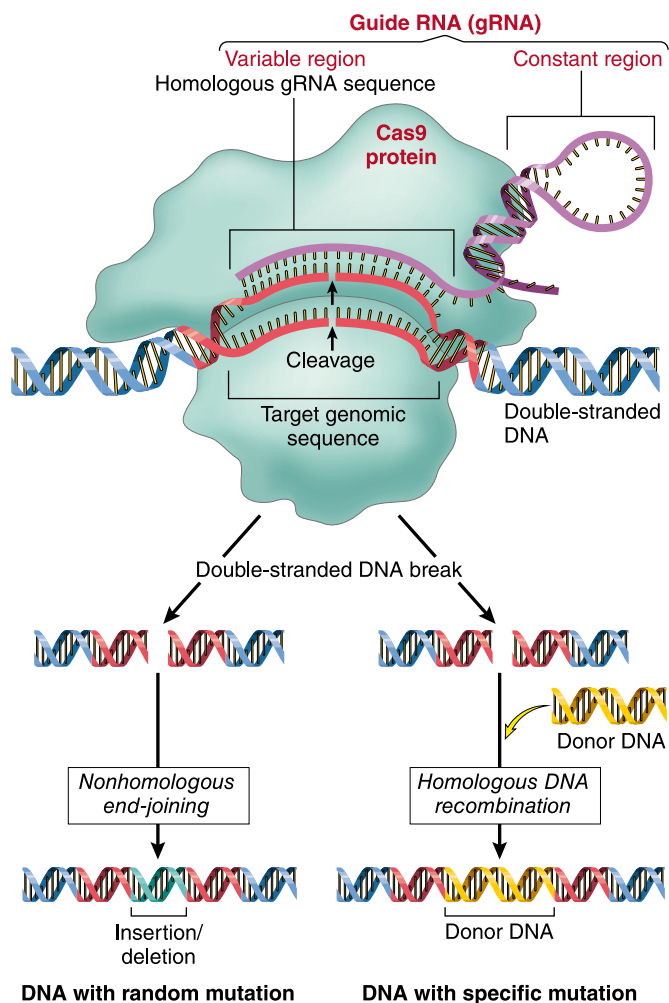


Fig. 1.5 Gene editing with clustered regularly interspaced short palindromic repeats (CRISPRs) and the nuclease Cas9. In bacteria, DNA sequences consisting of CRISPRs are transcribed into guide RNAs (gRNAs) with a constant region and a variable sequence of approximately 20 bases. The gRNA constant regions bind to Cas9, while the variable regions form heteroduplexes with homologous DNA sequences of interest; the Cas9 nuclease then cleaves the bound DNA to produce a double-stranded DNA break. To perform gene editing, gRNAs are designed with variable regions that are homologous to a specific DNA sequence of interest; co-expression of the gRNA and Cas9 then leads to efficient and highly specific cleavage of the target sequence. In the absence of homologous DNA, the double-stranded break is repaired by nonhomologous end joining (NHEJ), an error-prone mechanism that typically introduces disruptive insertions or deletions (indels). Conversely, in the presence of homologous “donor” DNA that spans the region targeted by the CRISPR/Cas9 complex, cells instead can use homologous DNA recombination (HDR) to repair the break. HDR is less efficient than NHEJ but has the capacity to introduce precise changes in DNA sequence. Potential applications of CRISPR/Cas9 coupled with HDR include repair of inherited genetic diseases and the creation of pathogenic mutations in inducible pluripotent stem cells for purposes of modeling various diseases.

Intracellular Organelles

Many of the normal housekeeping functions of the cell are compartmentalized within membrane-bound intracellular organelles (Fig. 1.6). By isolating certain cellular functions within distinct compartments, potentially injurious degradative enzymes or toxic metabolites can be kept at usefully high concentrations without risking damage to more delicate

intracellular constituents. Moreover, compartmentalization allows the creation of unique intracellular environments (e.g., low pH or high calcium) that permit more efficient functioning of certain enzymes or metabolic pathways.

Cell growth and maintenance require a constant supply of both energy and the building blocks that are needed for synthesis of macromolecules. Mitochondria serve as an important source of metabolic intermediates needed for anabolic metabolism, are sites of synthesis of certain macromolecules (e.g., heme), and contain important sensors of cell damage that can initiate and regulate programmed cell death (e.g., apoptosis). Most of the adenosine triphosphate (ATP) that powers cells is generated via *mitochondrial oxidative phosphorylation*.

New proteins destined for the plasma membrane or secretion are physically assembled in the *rough endoplasmic reticulum (RER)* and *Golgi apparatus*; proteins intended for the cytosol are synthesized on free ribosomes. *Smooth endoplasmic reticulum (SER)* is used for steroid hormone and lipoprotein synthesis and modification of hydrophobic compounds into water-soluble molecules for export.

Biosynthesis in cells is tightly coupled with catabolism to maintain homeostasis. Cells degrade the wide variety of molecules that they endocytose, as well as the entire repertoire of their own proteins and organelles—all of which are constantly being turned over. Breakdown of these constituents takes place at three different sites, ultimately serving different functions.

- *Proteasomes* are “disposal” complexes that degrade denatured or otherwise “tagged” cytosolic proteins.
- *Lysosomes* are intracellular organelles containing degradative enzymes that permit digestion of a wide range of macromolecules, including proteins, polysaccharides, lipids, and nucleic acids. They are the site of intracellular organelle breakdown (a process called *autophagy*) and where phagocytosed microbes are killed and catabolized.
- *Peroxisomes* contain catalase, peroxidase, and other oxidative enzymes; they play a specialized role in the breakdown of very long-chain fatty acids, generating hydrogen peroxide in the process.

The contents and location of cellular organelles are also highly regulated. *Endosomal vesicles* shuttle internalized material to the appropriate intracellular site(s), while other membrane-bound vesicles direct newly synthesized materials to the cell surface or specific organelles. Movement within the cell, as well as the entire cell in its environment, is accomplished by the *cytoskeleton*, which is composed of filamentous actin (microfilaments), keratins (intermediate filaments), and microtubules. These structural proteins also maintain cellular shape and intracellular organization, which are essential to generation and maintenance of *cell polarity*. This is particularly important in epithelium where the top of the cell (*apical*) and the bottom and sides of the cell (*basolateral*) are exposed to different environments and have distinct functions. Loss of polarity could, for example, disrupt vectorial transcellular transport in the intestine or renal tubule.

In growing and dividing cells, all these organelles have to be replicated (*organellar biogenesis*) and correctly apportioned in daughter cells following mitosis. Moreover, because the macromolecules and organelles have finite lifespans (mitochondria, e.g., last only about 10 days), mechanisms must also exist that allow for the recognition and degradation of “worn-out” cellular components.

In addition to the compartmentalization afforded by organelles, it is increasingly recognized that dynamic, membraneless compartments, called *biomolecular condensates*, govern important cellular housekeeping and gene regulatory functions in cells. These compartments consist of concentrated assemblies of proteins and nucleic acids within cells that form through a process called *phase separation*, resulting in condensed

Relative volumes of intracellular organelles (hepatocyte)

Compartment	% Total volume	Number/cell	Role in the cell
Cytosol	54%	1	metabolism, transport, protein translation
Mitochondria	22%	1700	energy generation, apoptosis
Rough ER	9%	1	synthesis of membrane and export protein
Smooth ER, Golgi	6%	1	protein modification, sorting, catabolism
Nucleus	6%	1	cell regulation, proliferation, DNA transcription
Endosomes	1%	200	intracellular transport and export
Lysosomes	1%	300	cellular catabolism
Peroxisomes	1%	400	very-long-chain fatty acid metabolism

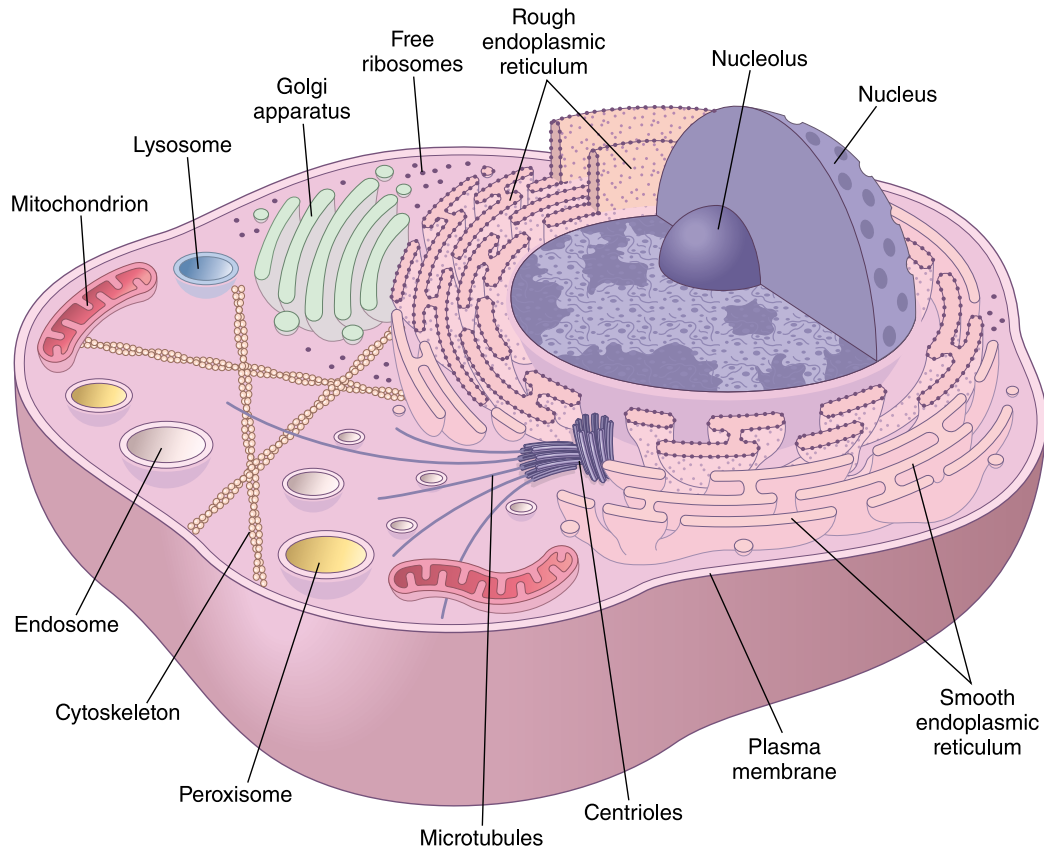


Fig. 1.6 Basic subcellular constituents of cells. ER, Endoplasmic reticulum. (Modified from Weibel ER, Stäubli W, Gnängi HR, et al. Correlated morphometric and biochemical studies on the liver cell. I. Morphometric model, stereologic methods, and normal morphometric data for rat liver. *J Cell Biol.* 1969;42(1):68–91.)

droplets. These granules separate out from the cytoplasm and form droplets, as oil does from water. Notable examples of biomolecular condensates include *stress granules*, *processing bodies (P-bodies)*, and *nucleoli*.

With this as a primer, we will now move on to discuss cellular components and their function in greater detail.

Plasma Membrane: Protection and Nutrient Acquisition

Plasma membranes (and all other organellar membranes) are more than just static lipid sheaths. Rather, they are fluid bilayers of amphipathic phospholipids—hydrophilic head groups that face the aqueous environment and hydrophobic lipid tails that interact with each other to form a barrier to passive diffusion of large or charged molecules (Fig. 1.7). The bilayer has a remarkably heterogeneous composition of different phospholipids that vary by location and are also asymmetric—that is, particular lipids are preferentially found within the extracellular or cytoplasmic faces of the bilayer. Proper localization of these molecules is important for cell health. For example, specific

phospholipids interact with particular membrane proteins and modify their distributions and functions.

- *Phosphatidylinositol* in the inner membrane leaflet can be phosphorylated, serving as an electrostatic scaffold for intracellular proteins; alternatively, polyphosphoinositides can be hydrolyzed by phospholipase C to generate intracellular second signals like diacylglycerol and inositol trisphosphate.
- *Phosphatidylserine* is normally restricted to the inner face where it confers a negative charge involved in electrostatic protein interactions; however, when flipped to the extracellular leaflet, it becomes a potent “eat me” signal during programmed cell death (e.g., apoptosis). In platelets, phosphatidylserine is also a cofactor in blood clotting.
- *Glycolipids* and *sphingomyelin* are preferentially located in the extracellular face; glycolipids, including gangliosides with complex sugar linkages and terminal sialic acids that confer negative charges, support charge-based interactions that contribute to inflammatory cell recruitment and sperm-egg fusion.

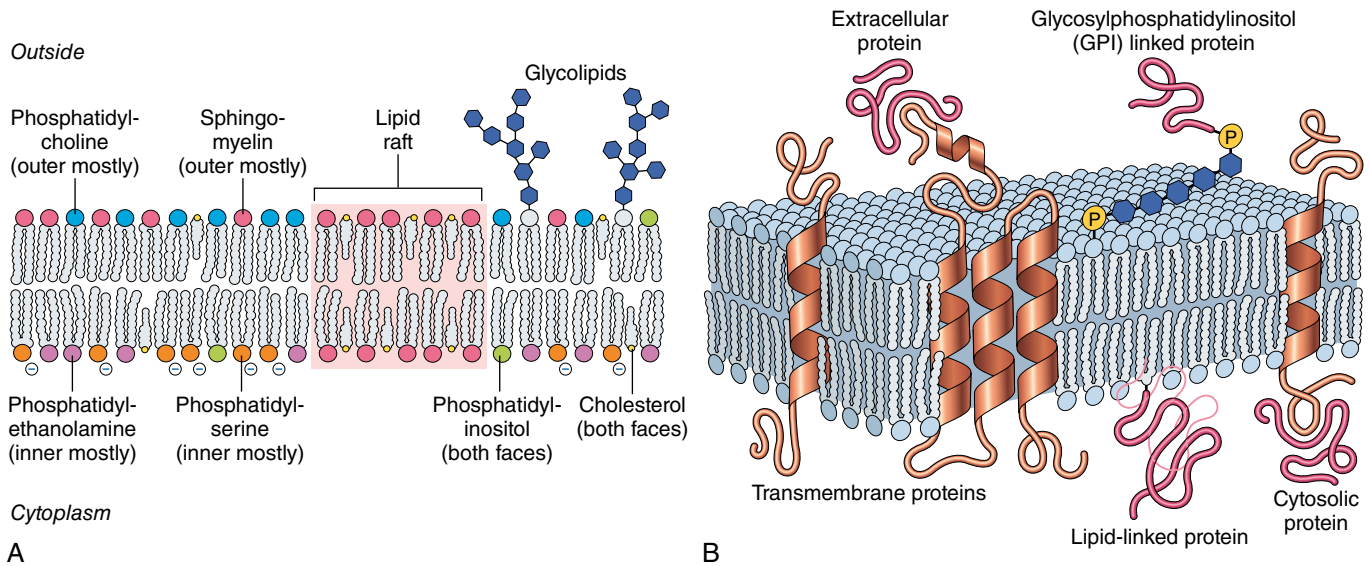


Fig. 1.7 Plasma membrane organization and asymmetry. (A) The plasma membrane is a bilayer of phospholipids, cholesterol, and associated proteins. The phospholipid distribution within the membrane is asymmetric due to the activity of flippases, transporter proteins that move specific phospholipids from one face of the bilayer to the other. As a result, phosphatidylcholine and sphingomyelin are overrepresented in the outer leaflet, and phosphatidylserine (negative charge) and phosphatidylethanolamine are predominantly found on the inner leaflet; glycolipids occur only on the outer face where they contribute to the extracellular glycocalyx. Although the membrane is laterally fluid and the various constituents can diffuse randomly, specific domains (e.g., cholesterol and glycosphingolipid-rich lipid rafts) can also form. (B) Membrane-associated proteins may traverse the membrane (singly or multiply) via α -helical hydrophobic amino acid sequences; depending on the membrane lipid content and relative hydrophobicity of protein domains, such proteins may have nonrandom distributions within the membrane. Proteins on the cytosolic face can be associated with the plasma membrane through posttranslational lipid modifications (e.g., farnesylation). Proteins on the extracytoplasmic face can associate with the membrane via glycosylphosphatidylinositol linkages. Transmembrane proteins can translate mechanical forces (e.g., from the cytoskeleton or extracellular matrix), as well as chemical signals across the membrane.

Despite substantial lateral fluidity, some membrane constituents concentrate into specialized domains (e.g., *lipid rafts*) that are enriched in glycosphingolipids and cholesterol. Since inserted membrane proteins have different intrinsic solubilities in domains with distinct lipid compositions, this membrane organization influences protein distribution. This geographic organization of plasma membrane components impacts cell-cell and cell-matrix interactions, intracellular signaling, and the specialized sites of vesicle budding or fusion.

The plasma membrane is liberally studded with a variety of proteins and glycoproteins involved in (1) ion and metabolite transport; (2) fluid-phase and receptor-mediated uptake of macromolecules; and (3) cell-ligand, cell-matrix, and cell-cell interactions. The means by which these proteins associate with membranes frequently reflects their function. For example, multiple transmembrane-spanning proteins are often pores or molecular transporters, while proteins that are superficially attached to the membrane via labile linkages are more likely to participate in signaling. In general, proteins associate with the lipid bilayer in one of four ways.

- As integral or transmembrane proteins, having one or more relatively hydrophobic α -helical segments that traverse the lipid bilayer.
- Through posttranslational addition of prenyl groups (e.g., farnesyl, related to cholesterol) or fatty acids (e.g., palmitic or myristic acid) that insert into the cytosolic side of the plasma membrane.

- Through glycosylphosphatidylinositol (GPI) tails that are added posttranslationally, a linkage used to associate extracellular proteins with the membrane.
- Indirectly, through noncovalent interactions with true transmembrane proteins.

Many plasma membrane proteins function as large complexes; these may assemble under the control of chaperone molecules in the RER or by lateral diffusion in the plasma membrane, followed by complex formation *in situ*. For example, many protein receptors (e.g., cytokine receptors) dimerize or trimerize in the presence of ligand to form functional signaling units. Although lipid bilayers are fluid within the plane of the membrane, components can be confined to discrete domains. This can occur by localization to lipid rafts (discussed earlier) or through intercellular protein-protein interactions (e.g., *tight junctions*) that establish discrete boundaries and have unique lipid composition. The latter strategy is used to maintain *cell polarity* in epithelial cells, which have distinct apical and basolateral membrane domains. Interactions of other membrane and cytosolic proteins with one another and the cytoskeleton also contribute to cell polarity.

The extracellular face of the plasma membrane is diffusely decorated by carbohydrates, not only as complex oligosaccharides on glycoproteins and glycolipids, but also as polysaccharide chains attached to integral membrane proteoglycans. This *glycocalyx* can form a chemical and mechanical barrier.

Membrane Transport

Transport of selected molecules across the lipid bilayer or to intracellular sites via vesicular transport is essential for cellular function. Several mechanisms contribute to this transport.

Passive Diffusion

Small, nonpolar molecules like O_2 and CO_2 readily dissolve in lipid bilayers and rapidly diffuse across them. Larger hydrophobic molecules, (e.g., steroid-based molecules like estradiol or vitamin D) can also cross lipid bilayers with relative ease. While small polar molecules such as water (18 Da) can also diffuse across membranes at low rates, in tissues responsible for regulating water movement (e.g., renal tubular epithelium), special integral membrane proteins called *aquaporins* form

transmembrane channels for water, H_2O_2 , and other small molecules. In contrast, the lipid bilayer is an effective barrier to the passage of larger polar molecules (>75 Da); for example, glucose (180 Da) is effectively excluded. Lipid bilayers are also impermeant to ions due to their charge and hydration.

Carriers and Channels

Plasma membrane transport proteins are responsible for uptake and secretion of ions and larger molecules that are required for cellular function (e.g., nutrient uptake and waste disposal). Ions and small molecules can be transported by channel proteins and carrier proteins (Fig. 1.8). Similar pores and channels also mediate transport across organellar membranes. The transporters that move ions, sugars, nucleotides,

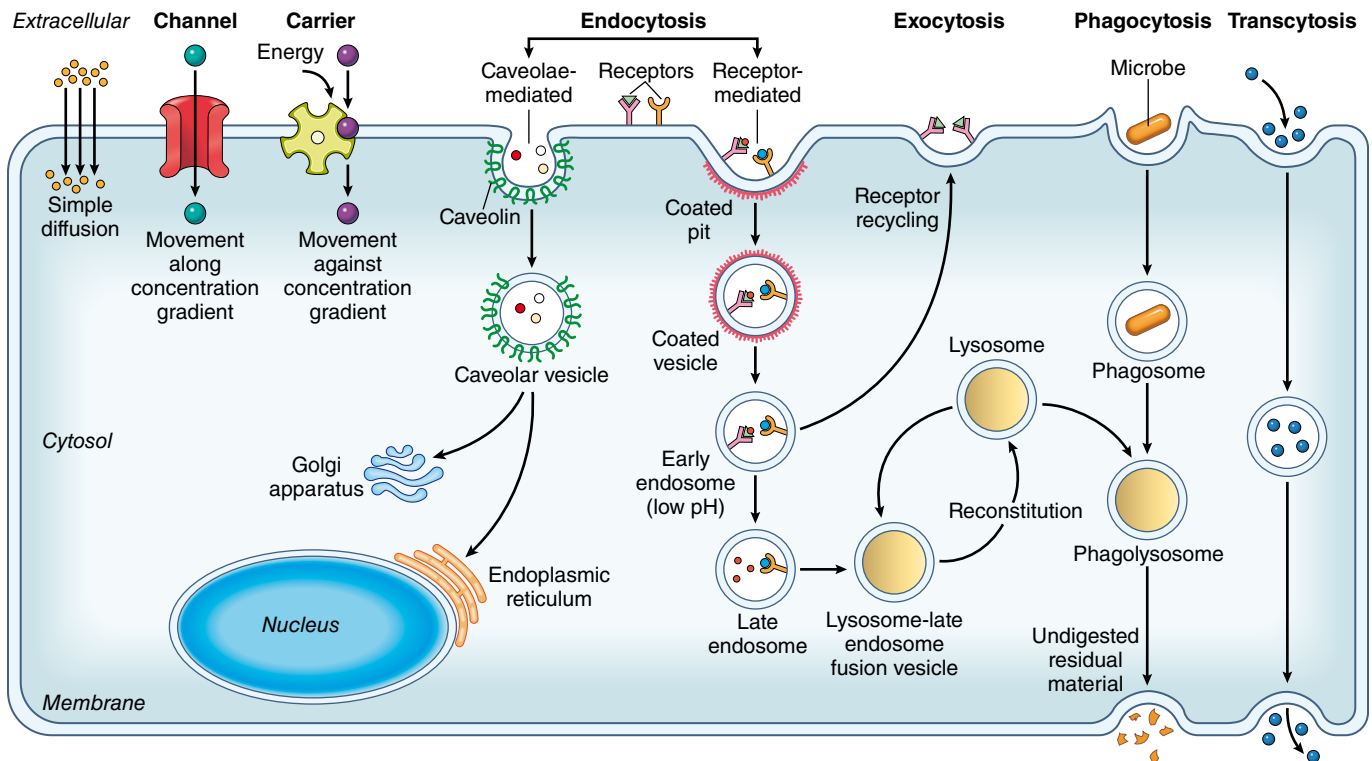


Fig. 1.8 Movement of small molecules and larger structures across membranes. The lipid bilayer is relatively impermeable to all but the smallest and/or most hydrophobic molecules. Thus, the import or export of charged species requires specific transmembrane transporter proteins, vesicular traffic, or membrane deformations. From left to right in the figure: Small charged solutes move across the membrane using either channels or carriers; in general, each molecule requires a unique transporter. Channels are used when concentration gradients drive solute movement, whereas carriers are required when solute is moved against a concentration gradient. Receptor-mediated and fluid-phase uptake of material involves membrane-bound vesicles. Caveolae endocytose extracellular fluid, membrane proteins, and some receptor-bound molecules (e.g., folate) in a process driven by caveolin proteins concentrated within lipid rafts. They subsequently fuse with endosomes or recycle to the membrane. Endocytosis of receptor-ligand pairs often involves clathrin-coated pits and vesicles. After internalization the clathrin disassembles and individual components can be reused. The resulting vesicle becomes part of the endocytic pathway, in which compartments are progressively more acidic. After ligand is released, the receptor can be recycled to the plasma membrane to repeat the process (e.g., iron dissociates from transferrin at pH ~ 5.5 ; apotransferrin and the transferrin receptor then return to the surface). Alternatively, receptor and ligand complexes can eventually be degraded within lysosomes (e.g., epidermal growth factor and its receptor are both degraded, which prevents excessive signaling). *Exocytosis* is the process by which membrane-bound vesicles fuse with the plasma membrane and discharge their contents to the extracellular space. This includes endosome recycling (shown), release of undigested residual material from lysosomes, transcytotic delivery of vesicles, and export of secretory vacuole contents (not shown). *Phagocytosis* involves membrane invagination to engulf large particles and is most common in specialized phagocytes (e.g., macrophages and neutrophils). The resulting phagosomes eventually fuse with lysosomes to facilitate the degradation of the internalized material. *Transcytosis* can mediate transcellular transport in either apical-to-basal or basal-to-apical directions, depending on the receptor and ligand.

and so forth, frequently have exquisite specificities and can be either active or passive (see below). For example, some transporters accommodate glucose but not galactose.

- *Channel proteins* create hydrophilic pores, which, upon opening, permit rapid movement of solutes (usually restricted by size and charge).
- *Carrier proteins* bind their specific solute and undergo a series of conformational changes to transfer the ligand across the membrane; their transport is relatively slow.

Solute transport across the plasma membrane is frequently driven by a concentration and/or electrical gradient between the inside and outside of the cell via *passive transport* (virtually all plasma membranes have an electrical potential difference across them, with the inside negative relative to the outside). In other cases, *active transport* of certain solutes (against a concentration gradient) is accomplished by carrier molecules (never channels) at the expense of ATP hydrolysis or a coupled ion gradient. For example, most apical nutrient transporters in the intestines and renal tubules exploit the extracellular to intracellular Na^+ gradient to allow absorption even when intracellular nutrient concentrations exceed extracellular concentrations. This form of active transport does not use ATP directly, but depends on the Na^+ gradient generated by Na^+, K^+ -ATPase. Other transporters are ATPases. One example is the *multidrug resistance (MDR) protein*, which pumps polar compounds (e.g., chemotherapeutic drugs) out of cells and may render cancer cells resistant to treatment.

Water movement into or out of cells is passive and directed by solute concentrations. Thus, extracellular salt in excess of that in the cytoplasm (*hypertonicity*) causes net movement of water out of cells, while *hypotonicity* causes net movement of water into cells. Conversely, the charged metabolites and proteins within the cytoplasm attract charged counterions that increase intracellular osmolarity. Thus, to prevent overhydration, cells must constantly pump out small inorganic ions (e.g., Na^+)—typically through the activity of the membrane ion-exchanging ATPase. Loss of the ability to generate energy (e.g., in a cell injured by toxins or ischemia) therefore results in osmotic swelling and eventual cell rupture. Similar transport mechanisms also regulate concentrations of other ions (e.g., Ca^{2+} and H^+). This is critical to many processes. For example, cytosolic enzymes are most active at pH 7.4 and are often regulated by Ca^{2+} , whereas lysosomal enzymes function best at pH 5 or less.

Uptake of fluids or macromolecules by the cell is called *endocytosis*. Depending on the size of the vesicle, endocytosis may be denoted pinocytosis (“cellular drinking”) or phagocytosis (“cellular eating”). Generally, phagocytosis is restricted to certain innate immune cells (e.g., macrophages and neutrophils) whose function depends on the ability to ingest invading organisms or dead cell fragments (Chapter 6).

Receptor-Mediated and Fluid-Phase Uptake

Certain small molecules—including some vitamins—bind to cell-surface receptors and are taken up through invaginations of the plasma membrane called *caveolae*. Uptake can also occur through membrane invaginations coated by an intracellular matrix of clathrin proteins that spontaneously assemble into a basket-like lattice, which helps drive endocytosis (discussed more later). In both cases, the activity of dynamin, a GTPase that acts as a molecular scissor, is required for vesicle release.

Macromolecules can also be exported from cells by *exocytosis*. In this process, proteins synthesized and packaged within the RER and Golgi apparatus are concentrated in secretory vesicles, which then fuse with the plasma membrane to expel their contents. Common examples include peptide hormones (e.g., insulin) and cytokines.

Transcytosis is the movement of endocytosed vesicles between the apical and basolateral compartments of cells. This is a mechanism for

transferring large amounts of intact proteins across epithelial barriers (e.g., ingested antibodies in maternal milk) or for rapid movement of large solute volumes.

We now return to the specifics of endocytosis (see Fig. 1.8).

- *Caveolae-mediated endocytosis*. *Caveolae* (“little caves”) are non-coated plasma membrane invaginations associated with GPI-linked molecules, cyclic adenosine monophosphate (cAMP) binding proteins, src-family kinases, and the folate receptor; *caveolin* is the major structural protein of caveolae, which, like membrane rafts (see earlier), are enriched in glycosphingolipids and cholesterol. Internalization of caveolae along with bound molecules and associated extracellular fluid is called *potocytosis*—literally “cellular sipping.” In addition to supporting transmembrane delivery of some molecules (e.g., folate), caveolae regulate transmembrane signaling and cellular adhesion via internalization of receptors and integrins.
- *Receptor-mediated endocytosis*. Macromolecules bound to membrane receptors (such as transferrin or low-density lipoprotein [LDL] receptors) are taken up at specialized regions of the plasma membrane called *clathrin-coated pits*. The receptors are efficiently internalized by membrane invaginations driven by the associated clathrin matrix, eventually pinching off to form *clathrin-coated vesicles*. Trapped within these vesicles is also a gulp of the extracellular milieu (*fluid-phase pinocytosis*). The vesicles then rapidly lose their clathrin coating and fuse with an acidic intracellular structure called the *early endosome*; the endosomal vesicles undergo progressive maturation to late endosomes, ultimately fusing with lysosomes. In the acidic environment of the endosomes, LDL and transferrin receptors release their cargo (cholesterol and iron, respectively), which is then transported into the cytosol.

After release of bound ligand, some receptors recycle to the plasma membrane and are reused (e.g., transferrin and LDL receptors), while others are degraded within lysosomes (e.g., epidermal growth factor receptor). In the latter case, degradation after internalization results in receptor downregulation that limits receptor-mediated signaling. Defects in receptor-mediated transport of LDL underlie familial hypercholesterolemia, as described in Chapter 5.

Endocytosis requires recycling of internalized vesicles back to the plasma membrane (exocytosis) for another round of ingestion. This is critical, as a cell will typically ingest from the extracellular space the equivalent of 10% to 20% of its own cell volume each hour—amounting to 1% to 2% of its plasma membrane each minute! Without recycling, the plasma membrane would be rapidly depleted. Endocytosis and exocytosis must therefore be tightly coupled to avoid large changes in plasma membrane area.

Cytoskeleton

The ability of cells to adopt a particular shape, maintain polarity, organize intracellular organelles, and migrate depends on an intracellular scaffold of structural proteins that form the cytoskeleton (Fig. 1.9). Although the cytoskeleton can have an appearance similar to the beams and supports of a building, cytoskeletal structures are constantly elongating and shrinking; microfilaments and microtubules are most active, and their assembly and disassembly can drive cell migration.

In eukaryotic cells, there are three major classes of cytoskeletal proteins.

- *Actin microfilaments* are 5- to 9-nm diameter fibrils formed from the globular protein actin (G-actin), the most abundant cytosolic protein in cells. G-actin monomers noncovalently polymerize into long filaments (F-actin) that intertwine to form double-stranded helices with a defined polarity. Although the details are (as always)

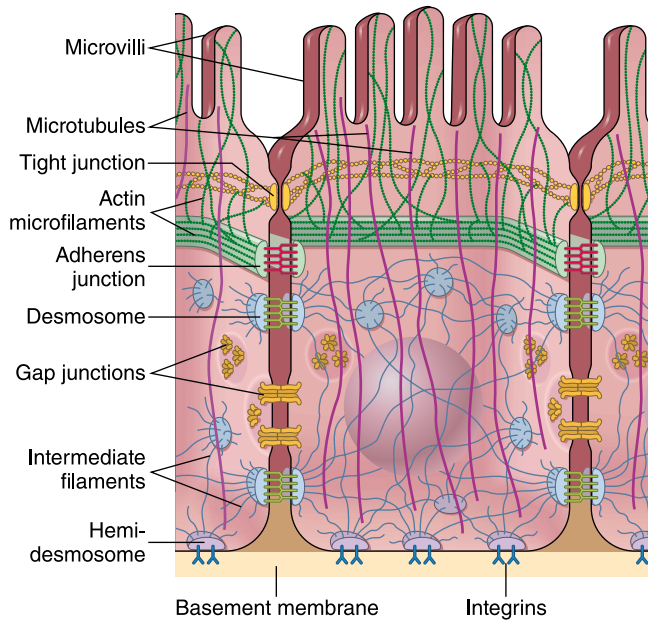


Fig. 1.9 Cytoskeletal elements and cell-cell interactions. Interepithelial adhesion involves several different protein-protein interactions within tight junctions, adherens junctions, and desmosomes, whereas adhesion to the extracellular matrix involves integrins (and associated proteins) within hemidesmosomes. The intracellular portions of adhesion proteins associate with actin microfilaments and intermediate filaments to create a protein matrix that maintains cellular structure and participates in intracellular signaling. Gap junctions do not impart structural integrity but allow cell-cell communication by the movement of small-molecular-weight and/or charged species. See text for details.

more nuanced, new subunits are typically added at the “positive” end of the strand and removed from the “negative” end—a process referred to as *actin treadmilling*. Actin nucleating, binding, and regulatory proteins organize polymerization, bundling, and branching to form networks that control cell shape and movement. This complex and its association with motor proteins (e.g., myosin) are so precisely arrayed in skeletal and cardiac muscle that a banding pattern is apparent by light microscopy. ATP hydrolysis by myosin slides the actin filaments relative to one another to cause muscle contraction. Although less coordinated, myosins, of which there are many, are responsible for other functions that depend on actin contraction, including vesicular transport, epithelial barrier regulation, and cell migration.

- **Intermediate filaments** are 10-nm diameter fibrils that comprise a large, heterogeneous family that includes keratin proteins and nuclear lamins. Intermediate filaments predominantly form ropelike polymers and do not usually actively reorganize like actin and microtubules. This allows intermediate filaments to provide tensile strength so that cells can bear mechanical stress, for example, in epithelia where intermediate filaments link *desmosomes* and *hemidesmosomes* (see Fig. 1.9). Individual intermediate filament proteins have characteristic tissue-specific patterns of expression that can be useful for assigning a cell of origin to poorly differentiated tumors. Examples include:
 - *Vimentin* in mesenchymal cells (fibroblasts, endothelium).
 - *Desmin* in muscle cells, which form a scaffold on which actin and myosin contract.
 - *Neurofilaments* in neurons are critical for neuronal axon structure and confer both strength and rigidity.

- *Glial fibrillary acidic protein* is expressed in glial cells.
- *Cytokeratins* are expressed in epithelial cells. There are at least 30 distinct different cytokeratins that are expressed in different cell lineages (e.g., lung vs. gastrointestinal epithelium).
- *Lamins* are intermediate filament proteins that form the nuclear lamina, define nuclear shape, and contribute to gene regulation.
- **Microtubules** are 25-nm-thick fibrils composed of noncovalently polymerized α - and β -tubulin dimers organized into hollow tubes. These fibrils are polarized, with “+” and “−” ends. The “−” end is typically embedded in a *microtubule organizing center* (MTOC or centrosome) near the nucleus, where it is associated with paired centrioles; the “+” end elongates or recedes in response to various stimuli by the addition or subtraction of tubulin dimers, making microtubules extremely dynamic. Microtubules:
 - Serve as mooring lines for molecular motor proteins that use ATP to translocate vesicles, organelles, or other molecules in cells. There are two varieties of motor proteins, kinesins and dyneins, that typically (but not exclusively) transport cargo in anterograde (− to +) or retrograde (+ to −) directions, respectively.
 - Mediate sister chromatid segregation during mitosis.
 - Form the core of *primary cilia*, single nonmotile projections on many nucleated cells that contribute to the regulation of cellular proliferation and differentiation (mutations in the proteins of the primary cilia complex lead to forms of polycystic kidney disease; see Chapter 20).
 - Can be adapted to form the core of motile cilia (e.g., in bronchial epithelium) or flagella (in sperm)

Cell-Cell Interactions

Cells connect and communicate with each other via junctional complexes that form mechanical links and facilitate receptor-ligand interactions. Similar complexes also mediate interaction with the ECM. Cell-cell junctions are organized into three basic types (see Fig. 1.9):

- **Ocluding junctions** (*tight junctions*) seal adjacent epithelial cells together to create a continuous barrier that restricts the paracellular (between cells) movement of ions and other molecules. Occluding junctions form a tight meshlike network (when viewed en face by freeze-fracture electron microscopy) of macromolecular contacts between neighboring cells. The complexes that mediate the cell-cell interactions are composed of transmembrane proteins including members of the *claudin* and *tight junction-associated MARVEL protein* (TAMP) tetraspan protein families. These connect to a host of intracellular adaptor and scaffolding proteins, including the three members of the *zonula occludens protein family* (ZO-1, ZO-2, ZO-3) and cingulin. Besides forming a selectively permeable barrier that seals the space between cells (i.e., the paracellular space), this zone also represents the boundary that separates apical and basolateral membrane domains and helps to maintain cellular polarity. However, these junctions are dynamic structures that can be modified to facilitate epithelial healing and inflammatory cell migration across epithelial-lined mucosal surfaces.
- **Anchoring junctions** (*adherens junctions* and *desmosomes*) mechanically attach cells—and their cytoskeletons—to other cells or the ECM. Adherens junctions are often closely associated with and beneath tight junctions. Desmosomes are more basal and form several types of junctions. When desmosomes attach the cell to the ECM they are referred to as hemidesmosomes (half a desmosome), since the other half of the desmosome is not present within the ECM. Both adherens junctions and desmosomes are formed by

homotypic extracellular interactions between transmembrane glycoproteins called *cadherins* on adjacent cells.

- In adherens junctions the transmembrane adhesion molecules are associated with intracellular actin microfilaments, through which they can also influence cell shape and/or motility. Loss of the epithelial adherens junction protein E-cadherin explains the discohesive invasion pattern of some gastric cancers and lobular carcinomas of the breast (see Chapters 17 and 23).
- In desmosomes the cadherins are linked to intracellular intermediate filaments, allowing extracellular forces to be mechanically communicated (and dissipated) over multiple cells.
- In hemidesmosomes the transmembrane connector proteins are *integrins*; like desmosomal cadherins, these attach to intermediate filaments and link the cytoskeleton to the ECM. *Focal adhesion complexes* are composed of >100 proteins and localize at hemidesmosomes. Their component proteins can generate intracellular signals when cells are subjected to shear stress (e.g., endothelium in the bloodstream or cardiac myocytes in a failing heart).
- *Communicating junctions (gap junctions)* permit the diffusion of chemical or electrical signals from one cell to another. The junction consists of a dense planar array of 1.5- to 2-nm pores (called *connexons*) formed by a pair of hexamers (one on each cell) of transmembrane connexin proteins. These form pores that permit passage of ions, nucleotides, sugars, amino acids, vitamins, and other small molecules. Permeability of the junction is rapidly reduced by lowered intracellular pH or increased intracellular calcium. Gap junctions play a critical role in cell-cell communication. For example, gap junctions in cardiac myocytes allow cell-to-cell calcium fluxes that permit the many cells of the myocardium to behave as a functional syncytium with coordinated waves of contraction.

Biosynthesis and Secretion: Endoplasmic Reticulum and Golgi Apparatus

All cellular constituents—including structural proteins, enzymes, transcription factors, and even the phospholipid membranes—are constantly renewed in an ongoing process balancing synthesis and degradation. The endoplasmic reticulum (ER) is the site of synthesis of all transmembrane proteins and lipids found in the plasma membrane and cellular organelles, including the ER itself. It is also the initial site of synthesis of secreted proteins. The ER is organized into a meshlike interconnected maze of branching tubes and flattened lamellae, forming a continuous sheet around a single lumen that is topologically equivalent to the extracellular environment. The ER is composed of contiguous but distinct domains that are distinguished by the presence (rough ER [RER]) or absence (smooth ER [SER]) of ribosomes (see Fig. 1.6).

- **RER:** Membrane-bound ribosomes on the cytosolic face of the RER translate mRNA into proteins that are extruded into the ER lumen or become integrated into the ER membrane. This process is directed by specific signal sequences on the N-termini of nascent proteins. Synthesis of proteins with signal peptides is initiated on free ribosomes, but the complex then becomes attached to the ER membrane, and the protein is inserted into or passed across the ER membrane as it is translated. For proteins lacking a signal sequence, translation remains on free ribosomes in the cytosol, forming polyribosomes as multiple ribosomes attach to the mRNA; such transcribed proteins remain within the cytoplasm.
- Proteins inserted into the ER fold into their active conformation and can form polypeptide complexes (oligomerize); in addition, disulfide bonds are formed, and N-linked oligosaccharides (sugar moieties attached to asparagine residues) are added. Chaperone molecules assist in folding and retaining proteins in the ER until

the modifications are complete and the proper conformation is achieved. If a protein fails to appropriately fold or oligomerize, it is retained and degraded within the ER. A good example of this is the most common mutation of the CFTR protein in cystic fibrosis. In this mutant CFTR, a codon deletion leads to the absence of a single amino acid (Phe508), which results in its misfolding, ER retention, and catabolism and therefore reduced surface expression. Moreover, excess accumulation of misfolded proteins—exceeding the capacity of the ER to edit and degrade them—leads to the ER stress response (also called the *unfolded protein response* [UPR]) (see Fig. 1.10B). Detection of excess abnormally folded proteins leads to a reduction in protein synthesis overall with a concurrent increase in chaperone proteins; failure to correct the overload can trigger cell death through apoptosis (Chapter 2).

- **Golgi apparatus:** From the RER, proteins and lipids destined for other organelles or extracellular export are shuttled into the Golgi apparatus. This consists of stacked cisternae that progressively modify proteins in an orderly fashion from *cis* (near the ER) to *trans* (near the plasma membrane). Cisternal progression (i.e., movement of *cis*-face cisternae to the *trans* aspect of the Golgi, akin to steps on an escalator) allows sequential processing of newly synthesized proteins and can be facilitated by small membrane-bound vesicles. Similar vesicles shuttle Golgi-resident enzymes from *trans* to *cis* to maintain the different cisternal contents along this assembly line. As cisternae mature, the N-linked oligosaccharides originally added in the ER are pruned and extended in a stepwise fashion; O-linked oligosaccharides (sugar moieties linked to serine or threonine) are also appended. Some of this glycosylation is important in sorting molecules to lysosomes (via the mannose-6-phosphate receptor); other glycosylation adducts are important for cell-cell or cell-matrix interactions or for clearing senescent cells (e.g., platelets and erythrocytes). In the *trans Golgi network*, proteins and lipids are sorted and dispatched to other organelles, plasma membrane, or secretory vesicles. The Golgi complex is especially prominent in cells specialized for secretion, including goblet cells of the intestinal or bronchial epithelium, which secrete large amounts of polysaccharide-rich mucus. In plasma cells that secrete antibodies, the Golgi can be recognized as a perinuclear hof on hematoxylin and eosin stains (Chapter 6).
- **SER:** In most cells, the SER is relatively sparse and primarily exists as the transition zone extending from RER to generate transport vesicles that carry newly synthesized proteins to the Golgi apparatus. The SER may, however, be particularly conspicuous in cells that synthesize steroid hormones (e.g., within the gonads or adrenals) or that catabolize lipid-soluble molecules (e.g., hepatocytes). Indeed, repeated exposure to compounds that are metabolized by the SER (e.g., phenobarbital, which is catabolized by the cytochrome P-450 system) can lead to SER hyperplasia. The SER is also responsible for sequestering intracellular calcium, which, when released into the cytosol, can mediate several responses to extracellular signals (including apoptotic cell death). In muscle cells, specialized SER called *sarcoplasmic reticulum* is responsible for the cyclic release and sequestration of calcium ions that regulate muscle contraction and relaxation, respectively.

The transport of proteins through ER and Golgi apparatus described earlier is termed the *classical secretory pathway*. However, certain molecules are released directly from inside a cell into the extracellular space through a collection of processes termed *unconventional secretion*. Some important examples of unconventional secretion include:

- Interleukin-1 β , a key proinflammatory cytokine, and α -synuclein, a protein implicated in Parkinson disease.

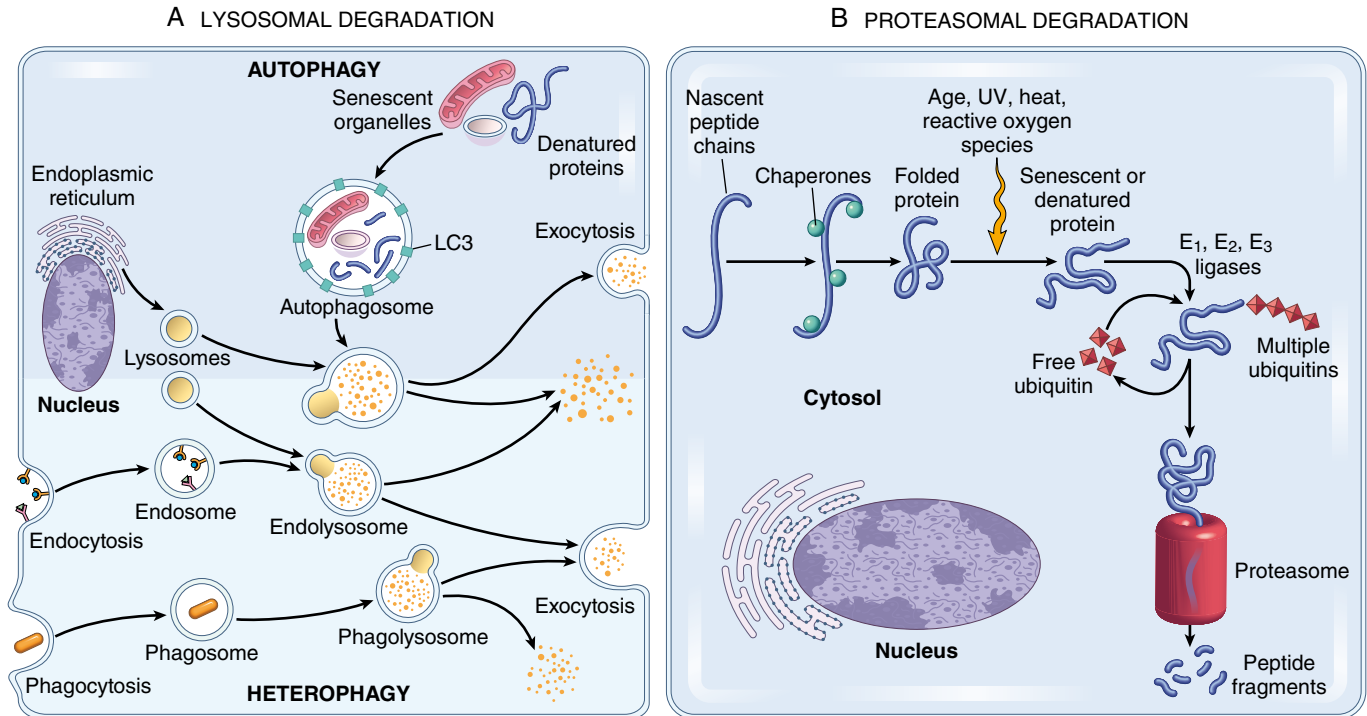


Fig. 1.10 Intracellular catabolism. (A) Lysosomal degradation. In autophagy (top of panel A), senescent organelles or denatured proteins are targeted for lysosome-driven degradation as they are encompassed within a double-membrane vacuole and marked by microtubule-associated protein 1 light chain 3 (LC3). Cell stress, such as nutrient depletion or some intracellular infections, can also activate the autophagic pathway. In heterophagy (bottom of panel A), lysosomes fuse with endosomes or phagosomes to facilitate the degradation of their internalized contents (see Fig. 1.8). The end products from both autophagy and heterophagy pathways may be released into the cytosol for nutrition or discharged into the extracellular space (lysosomal exocytosis). (B) Proteasomal degradation. Cytosolic proteins destined for turnover (e.g., transcription factors or regulatory proteins), senescent proteins, or proteins that have become denatured due to extrinsic mechanical or chemical stresses are tagged by multiple ubiquitin molecules (through the activity of E₁, E₂, and E₃ ubiquitin ligases). Ubiquitinated proteins are marked for degradation by proteasomes, cytosolic multi-subunit complexes that degrade proteins to small peptide fragments.

- **Extracellular vesicles (EVs)**, including *exosomes* and *microvesicles*, small membrane-bound nanovesicles containing proteins, lipids, and nucleic acids that are released into the extracellular space. EVs present in biofluids, particularly in blood, have gained attention as a source of tissue biomarkers that can be measured in the blood (so-called liquid biopsy).

Waste Disposal: Lysosomes and Proteasomes

Although cells rely primarily on lysosomes to digest internalized material and accumulated internal waste, there are multiple other routes to degrade intracellular macromolecules (see Fig. 1.10). These include proteasomes and peroxisomes. The latter are responsible for catabolism of long-chain fatty acids.

- **Lysosomes** are membrane-bound organelles containing roughly 40 different acid hydrolases that function best at pH ≤ 5; these include proteases, nucleases, lipases, glycosidases, phosphatases, and sulfatases. Lysosomal enzymes are initially synthesized in the ER lumen and then tagged with mannose-6-phosphate (M6P) within the Golgi apparatus. These M6P-modified proteins are subsequently delivered to lysosomes through *trans* Golgi vesicles that express M6P receptors. The other macromolecules destined for catabolism in the lysosomes arrive by one of three pathways (see Fig. 1.10).
 - Material internalized by *fluid-phase* or *receptor-mediated endocytosis* passes from plasma membrane to early and then late

endosomes and ultimately arrives at the lysosome. These compartments are progressively acidified such that proteolytic enzymes become active in late endosome and lysosomes.

- Damaged or superfluous organelles and/or large protein complexes can be ferried into lysosomes by a process called *autophagy*. Through a mechanism involving the products of a number of autophagy-related (*Atg*) genes, obsolete organelles are corralled by a double membrane derived from the ER. The membrane progressively expands to encircle a collection of organelles and cytosolic constituents, forming the definitive *autophagosome*; these structures are targeted for eventual destruction by fusion with lysosomes. In addition to facilitating turnover of aged and/or defunct structures, autophagy promotes nutrient salvage during starvation, participates in intracellular repair, and supports protective responses to intracellular infections. Autophagy is discussed in more detail in Chapter 2.
- **Phagocytosis** of microorganisms or large fragments of matrix or debris occurs primarily in professional phagocytes (macrophages or neutrophils). The material is engulfed to form a *phagosome* that subsequently fuses with lysosomes.
- **Proteasomes** play an important role in degrading cytosolic proteins (see Fig. 1.10); these include denatured or misfolded proteins as well as any other macromolecule whose lifespan needs to be regulated (e.g., signaling molecules). Many proteins destined for

destruction are first tagged covalently with a small protein called *ubiquitin*. Polyubiquitinated molecules are unfolded and funneled into the polymeric proteasome complex—a cylinder containing multiple protease activities, each with its active site pointed at the hollow core. Proteasomes digest proteins into small (6 to 12 amino acids) fragments that can subsequently be degraded to their constituent amino acids and recycled. In antigen-presenting cells, the resulting short peptides are presented in the context of class I or class II major histocompatibility molecules to help direct the adaptive immune response (Chapter 6). In other cases, proteasomal degradation of regulatory proteins or transcription factors can modulate signaling pathways.

CELLULAR METABOLISM AND MITOCHONDRIAL FUNCTION

Mitochondria evolved from ancestral prokaryotes that were engulfed by primitive eukaryotes about 1.5 billion years ago. Their origin explains why mitochondria contain their own DNA that encodes about 1% of total cellular protein and approximately 20% of the proteins involved in oxidative phosphorylation. Although their genomes are small, mitochondria can carry out all the steps of DNA replication, transcription, and translation using machinery similar to present-day bacteria; for example, mitochondria initiate protein synthesis with N-formylmethionine and are sensitive to some antibacterial antibiotics.

Since mitochondria biogenesis requires a genetic contribution from preexisting mitochondria and the ovum contributes the vast majority of cytoplasmic organelles in the fertilized zygote, mitochondrial DNA is almost entirely maternally inherited. Nevertheless, because the protein constituents of mitochondria are encoded by both nuclear and mitochondrial DNA, mitochondrial disorders may be X-linked, autosomal, or maternally inherited.

Mitochondria are impressively dynamic, constantly undergoing fission and fusion with other newly synthesized mitochondria; this supports their renewal and defends against degenerative changes that occur through ongoing oxygen free radical damage. Even so, mitochondria are short-lived—being degraded through autophagy (a process called *mitophagy*)—with estimated half-lives of 1 to 10 days, depending on the tissue, nutritional status, metabolic demands, and intercurrent injury.

Besides providing the enzymatic machinery for oxidative phosphorylation (and thus the relatively efficient generation of energy from glucose and fatty acid substrates), mitochondria play a fundamental role in regulating apoptosis (Fig. 1.11). Details of mitochondrial functions are:

- **Energy generation.** Each mitochondrion has two separate membranes with distinct functions. The inner membrane contains the enzymes of the respiratory chain folded into cristae. This encloses a core matrix space that harbors the bulk of the enzymes of the glycolytic and tricarboxylic acid cycles, and these constitute the major working parts of the organelle. Outside the inner membrane is the

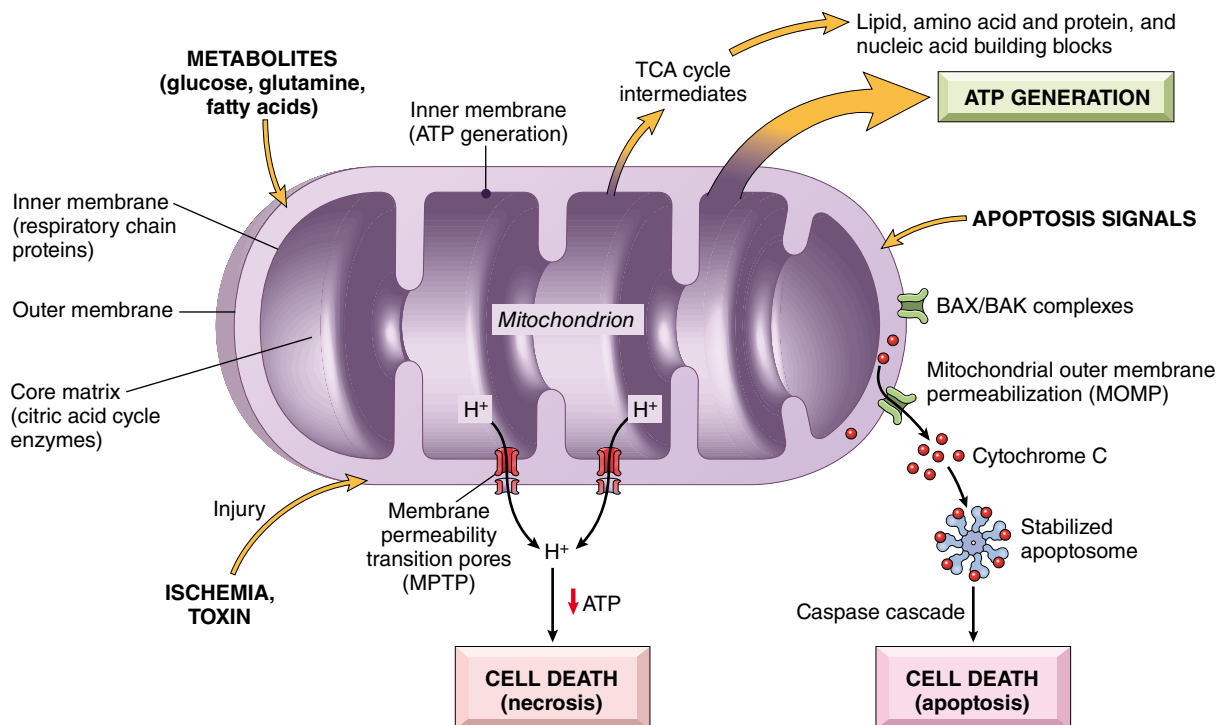


Fig. 1.11 Roles of the mitochondria. Besides the efficient generation of adenosine triphosphate (ATP) from carbohydrate and fatty acid substrates, mitochondria play an important role in intermediary metabolism—serving as the source of substrates used to synthesize lipids, proteins, and nucleotides. Significantly, they are centrally involved in cell life-and-death decisions: (1) toxic or ischemic injury will induce a membrane permeability transition that dissipates the intermembrane proton gradient, leading to cell death through loss of ATP generation; (2) intracellular signaling from both intrinsic and extrinsic sources can also result in the formation of oligomerized BAX and BAK protein pores resulting in MOMP that facilitates release of cytochrome C from the electron transport chain proteins. Cytosolic cytochrome C stabilizes the multi-subunit apoptosome to promote caspase activation and ultimately apoptotic cell death. TCA, Tricarboxylic acid.

intermembrane space, which is the site of nucleotide phosphorylation and is, in turn, enclosed by the outer membrane; the latter is studded with porin proteins that form voltage-dependent anion channels that are permeable to small (<5000 Da) molecules. As with other cellular membranes, larger molecules (and even some smaller polar species) require specific transporters.

- The major source of the energy that facilitates all cellular functions derives from oxidative metabolism. Mitochondria oxidize substrates to CO₂, transferring the high-energy electrons from the original molecule (e.g., sugar) to molecular oxygen. Oxidation of various metabolites drives proton pumps that transfer H⁺ from the core matrix into the intermembrane space. As the H⁺ ions flow down their electrochemical gradient and out of the intermembrane space, the energy released is used to generate ATP.
- Notably, the electron transport chain need not necessarily be coupled to ATP generation. An inner membrane protein enriched in brown fat called *thermogenin* (also called *uncoupling protein-1* [UCP-1]) is a hydrogen ion transporter that can dissipate the proton gradient, uncoupling it from oxidative phosphorylation. By this means, there is rapid substrate oxidation without ATP synthesis that allows tissues with high levels of UCP-1 to generate heat (nonshivering thermogenesis). As a natural (albeit usually low-level) by-product of substrate oxidation and electron transport, mitochondria are also an important source of reactive oxygen species (oxygen free radicals, hydrogen peroxide, etc.). Importantly, hypoxia, toxic injury, or even mitochondrial aging can lead to oxidative stress, characterized by increases in intracellular reactive oxygen species.
- *Oxidative phosphorylation.* Oxidative phosphorylation efficiently generates 36 to 38 ATP molecules per glucose molecule, but also “burns” substrates to their core CO₂ and H₂O, leaving no carbon moieties to use for building lipids and proteins. Consequently, to ensure the necessary building blocks for growth are available, rapidly dividing cells (both benign and malignant) increase their uptake of glucose and glutamine and switch to aerobic glycolysis, a phenomenon called the *Warburg effect*. In that setting, each glucose molecule is catabolized to lactic acid (even in the presence of adequate oxygen), generating only two net ATP molecules but “spinning off” intermediates that can be converted into new lipids, amino acids and proteins, and nucleic acids. Thus, mitochondrial metabolism can be modulated to support either cellular energy maintenance or cellular proliferation.
- *Cell death.* In addition to providing the ATP that enables the bulk of cellular activity, mitochondria are fundamental to cell survival. The role of mitochondria in cell death is detailed in Chapter 2 and briefly mentioned here.
 - *Necrosis:* External cellular injury (toxin, ischemia, trauma) can damage mitochondria, inducing the protracted opening of mitochondrial permeability transition pores (MPTP). These pores allow the dissipation of the electrochemical gradient so that subsequent mitochondrial ATP generation becomes impossible, and the cell dies.
 - *Apoptosis:* Programmed cell death is a central feature of normal tissue development and turnover and can be triggered by extrinsic signals (including cytotoxic T cells or inflammatory cytokines) or intrinsic pathways (including DNA damage or intracellular stress). Mitochondria integrate intracellular proapoptotic and antiapoptotic effector signals to generate a final “go” or “no go” signal for apoptosis. A net proapoptotic signal results in mitochondrial outer membrane permeabilization (MOMP)—distinct from the mitochondrial permeability transition—that releases

cytochrome C (and other proteins) into the cytoplasm. These proteins activate intracellular programmed cell death pathways. Notably, a failure of normal proapoptotic signaling (or an excess of antiapoptotic signaling) can underlie malignancy—even in the face of mutations that would otherwise trigger cellular suicide. Conversely, too strong an apoptotic signal (or lack of antiapoptotic effectors) may lead to premature cell death, as in neurodegenerative disorders (Chapter 28).

CELLULAR SIGNALING

Intercellular communication is essential to multicellular organisms. At the most basic level, extracellular signals may determine whether a cell lives or dies, remains quiescent, or becomes active to perform its specific functions. Intercellular signaling is critical in the developing embryo and in maintaining tissue organization, and is also important in adult organisms, where it assures that cells within tissues respond in a coordinated fashion (e.g., in response to local tissue trauma or a systemic infection). Loss of intercellular communication and “social controls” can be reflected in unregulated growth (e.g., cancer) or in detrimental responses to extrinsic stress (e.g., shock).

Individual cells are chronically exposed to a remarkable cacophony of signals that must be integrated into rational output; some signals may induce a given cell type to differentiate, while others induce proliferation or direct the cell to perform specialized functions. Many cells require certain level of stimulation to remain alive; in the absence of appropriate exogenous signals, they die by apoptosis.

Signaling pathways are generally classified based on the spatial relationships between the sending and receiving cells.

- *Paracrine signaling.* Only cells in the immediate vicinity are affected. To accomplish this, there can be only minimal diffusion, after which the secreted signal is rapidly degraded, taken up by other cells, or trapped in the ECM.
- *Autocrine signaling* occurs when molecules secreted by a cell affect that same cell. This can be a means to entrain groups of cells undergoing synchronous differentiation during development, or it can be used to amplify (positive feedback) or dampen (negative feedback) a particular response.
- *Endocrine signaling.* A mediator is released into the bloodstream and acts on target cells at a distance.
- *Synaptic signaling.* Activated neurons secrete neurotransmitters at specialized cell junctions (i.e., synapses) onto target cells.

The signals that most cells respond to can be classified into several groups.

- *Danger and pathogens.* Many cells have an innate capacity to sense and respond to damaged cells (danger signals), as well as foreign invaders such as microbes. The receptors that recognize these threats are discussed in Chapters 3 and 6.
- *Cell-cell contacts,* mediated through adhesion molecules and/or gap junctions. As mentioned previously, gap junction signaling is accomplished between adjacent cells via hydrophilic connexon channels that permit the movement of small ions (e.g., calcium), metabolites, and second messenger molecules (e.g., cAMP).
- *Cell-ECM contacts,* mediated through integrins. We will return to a consideration of integrins in the context of leukocyte attachment to other cells during inflammation in Chapter 3.
- *Secreted molecules.* The most important secreted molecules include growth factors (discussed later); cytokines, a term reserved for mediators of inflammation and immune responses (discussed also in Chapters 3 and 6); and hormones, which are secreted by endocrine organs (Chapter 24).

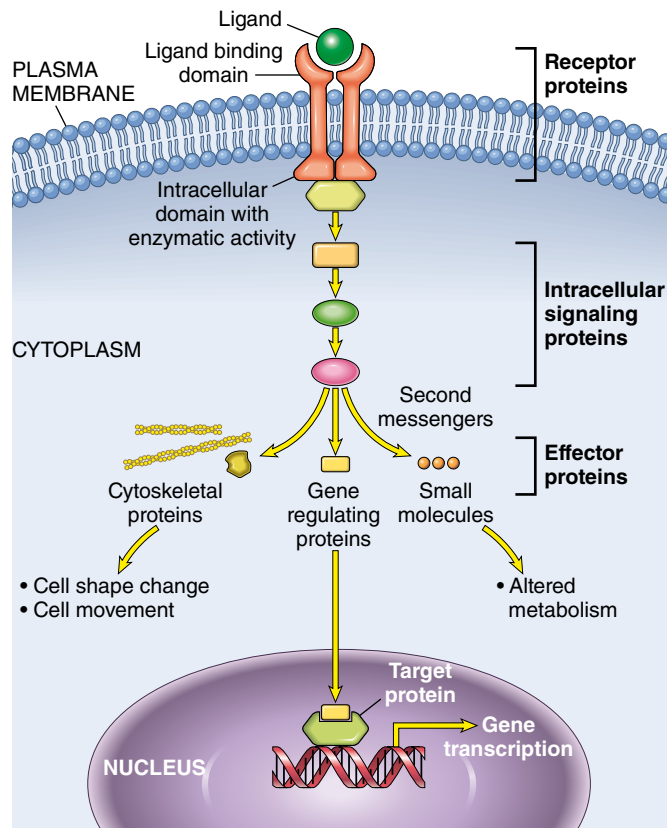


Fig. 1.12 Modular signaling from a receptor at the plasma membrane. The binding of a receptor to ligand in the extracellular milieu results in the activation of an intracellular enzymatic activity (e.g., phosphorylation, proteolytic cleavage), which triggers the propagation of a signal via one or more intracellular signaling adaptor proteins. Ultimately, these signaling cascades activate effector proteins that orchestrate the response to the initial cue through changes in fundamental cell biological activities, such as gene transcription, altered metabolism, and cytoskeletal-mediated shape changes and migration.

Regardless of the nature of an extracellular stimulus (paracrine, autocrine, synaptic, or endocrine), the signal that is conveyed is transmitted to the cell via specific receptor proteins. Signaling molecules (ligands) bind their respective receptors and initiate a cascade of intracellular events culminating in the desired cellular response (Fig. 1.12). Ligands usually have high affinities for their receptors ($\leq 10^{-8}$ M) and, at physiologic concentrations, bind receptors with exquisite specificity. Receptors may be present on the cell surface or within the cell.

- **Intracellular receptors** include transcription factors that are activated by lipid-soluble ligands that can easily transit plasma membranes; vitamin D and steroid hormones that activate nuclear hormone receptors are good examples. In other cases, a small and/or nonpolar signaling ligand produced by one cell type can influence the activity of adjacent cells. Thus, nitric oxide produced by endothelial cells diffuses into underlying smooth muscle cells, activating the enzyme guanylyl cyclase to generate cyclic guanosine monophosphate (cGMP) that causes smooth muscle relaxation; endothelium can thereby regulate vasomotor tone.
- **Cell-surface receptors** are generally transmembrane proteins with extracellular domains that bind activating ligands. Depending on the receptor, ligand binding can:
 - Open ion channels, typically at the synapse between electrically excitable cells.

- Activate an associated GTP-binding regulatory protein (G protein).
- Activate an endogenous or associated enzyme (often a tyrosine kinase).
- Trigger a proteolytic event or change protein binding or stability to activate a latent transcription factor.

G protein-coupled receptors and tyrosine kinase-associated receptors are typically involved in signaling that drives cellular proliferation; proteolytic or conformational changes are common features of multiple pathways (e.g., Notch, Wnt, and Hedgehog) that influence normal development. Secondary intracellular downstream events frequently involve phosphorylation or dephosphorylation of target molecules, with subsequent conformational changes that impact nuclear access or enzymatic activity (see later). Understandably, signals transduced by cell-surface receptors are frequently perturbed in developmental disorders and malignancy.

Signal Transduction Pathways

The interaction of a cell-surface receptor and its ligand can activate signaling through ligand-induced clustering of the receptor (receptor cross-linking) or by inducing a physical change in receptor structure (see Fig. 1.12). Either mechanism results in a conformational change in the cytosolic tail of the receptor that mediates additional intracellular biochemical events.

Cellular receptors are grouped into several types based on the signaling mechanisms they use and the intracellular biochemical pathways they activate (Fig. 1.13). Receptor signaling most commonly leads to the formation or modification of biochemical intermediates and/or activation of enzymes that ultimately lead to generation of active transcription factors that enter the nucleus and alter the pattern of gene expression.

- **Receptors associated with kinase activity.** Downstream phosphorylation is a common pathway of signal transduction. Changes in receptor geometry can stimulate intrinsic receptor protein kinase activity or promote the enzymatic activity of recruited intracellular kinases; these kinases add charged phosphate residues to target molecules. Tyrosine kinases phosphorylate specific tyrosine residues, whereas serine/threonine kinases add phosphates to serine or threonine residues, and lipid kinases phosphorylate lipid substrates. For every phosphorylation event, there is also a phosphatase that removes phosphate residues to modulate signaling; phosphatases usually inhibit signal transduction.
 - **Receptor tyrosine kinases (RTKs)** (e.g., receptors for insulin, epidermal growth factor, and platelet-derived growth factor) are integral membrane proteins; ligand-induced cross-linking activates intrinsic tyrosine kinase domains located in their cytoplasmic tails.
 - **Nonreceptor tyrosine kinases.** Several kinds of receptors have no intrinsic catalytic activity (e.g., immune receptors, some cytokine receptors, and integrins). For these, a separate intracellular protein—known as a *nonreceptor tyrosine kinase*—interacts with receptors after ligand binding and phosphorylates specific motifs on the receptor or other proteins. The cellular homologue of the transforming protein of the Rous sarcoma virus, called Src, is the prototype for an important family of such nonreceptor tyrosine kinases (*Src-family kinases*). Src contains unique functional regions called *Src homology* (SH) domains; SH2 domains typically bind to receptors phosphorylated by another kinase, allowing the aggregation of multiple enzymes, whereas SH3 domains mediate other protein-protein interactions, often involving proline-rich sequences.

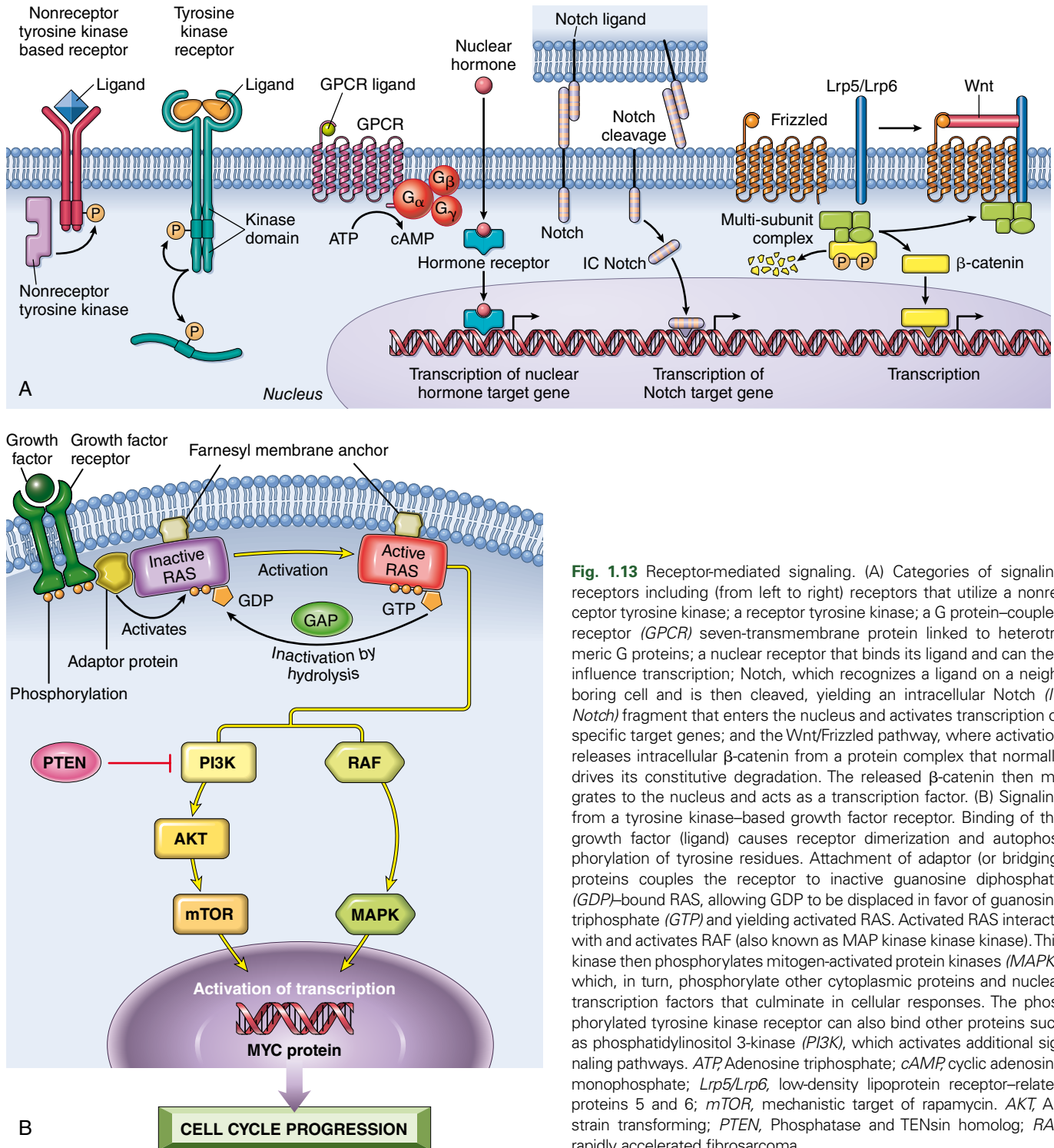


Fig. 1.13 Receptor-mediated signaling. (A) Categories of signaling receptors including (from left to right) receptors that utilize a nonreceptor tyrosine kinase; a receptor tyrosine kinase; a G protein-coupled receptor (GPCR) seven-transmembrane protein linked to heterotrimeric G proteins; a nuclear receptor that binds its ligand and can then influence transcription; Notch, which recognizes a ligand on a neighboring cell and is then cleaved, yielding an intracellular Notch (IC Notch) fragment that enters the nucleus and activates transcription of specific target genes; and the Wnt/Frizzled pathway, where activation releases intracellular β -catenin from a protein complex that normally drives its constitutive degradation. The released β -catenin then migrates to the nucleus and acts as a transcription factor. (B) Signaling from a tyrosine kinase-based growth factor receptor. Binding of the growth factor (ligand) causes receptor dimerization and autophosphorylation of tyrosine residues. Attachment of adaptor (or bridging) proteins couples the receptor to inactive guanosine diphosphate (GDP)-bound RAS, allowing GDP to be displaced in favor of guanosine triphosphate (GTP) and yielding activated RAS. Activated RAS interacts with and activates RAF (also known as MAP kinase kinase). This kinase then phosphorylates mitogen-activated protein kinases (MAPK), which, in turn, phosphorylate other cytoplasmic proteins and nuclear transcription factors that culminate in cellular responses. The phosphorylated tyrosine kinase receptor can also bind other proteins such as phosphatidylinositol 3-kinase (PI3K), which activates additional signaling pathways. *ATP*, Adenosine triphosphate; *cAMP*, cyclic adenosine monophosphate; *Lrp5/Lrp6*, low-density lipoprotein receptor-related proteins 5 and 6; *mTOR*, mechanistic target of rapamycin. *AKT*, AK strain transforming; *PTEN*, Phosphatase and TENSin homolog; *RAF*, rapidly accelerated fibrosarcoma.

- *G protein-coupled receptors* (GPCRs) characteristically traverse the plasma membrane seven times (hence their designation as seven-transmembrane or serpentine receptors); more than 1500 such receptors have been identified. After ligand binding, the GPCR associates with an intracellular guanosine triphosphate (GTP)-binding protein (G protein) that contains guanosine diphosphate (GDP). G protein interaction with a GPCR-ligand complex results in activation through the exchange of GDP for GTP. Subsequent GPCR-mediated signaling pathways include the generation of cAMP and

inositol-1,4,5-triphosphate (IP₃), with the latter triggering release of calcium from the ER.

- *Nuclear receptors*. Lipid-soluble ligands can diffuse into cells where they interact with intracellular proteins to form a receptor-ligand complex that directly binds to nuclear DNA; the results can be either activation or repression of gene transcription.
- *Other classes of receptors*. Other receptors—originally recognized as important for embryonic development and cell fate determination—are now recognized to participate in the functioning of mature cells,

particularly within the immune system. Rather than enzymatic activity, these pathways rely on protein-protein interactions to transduce signals.

- Receptor proteins of the *Notch* family fall in this category; ligand binding to Notch receptors leads to proteolytic cleavage of the receptor and subsequent nuclear translocation of its cytoplasmic domain (intracellular Notch) to form a transcription complex.
- *Wnt* protein ligands can also influence cell development through a canonical pathway involving transmembrane *Frizzled* family receptors, a distinct set of GPCRs that regulate intracellular levels of β -catenin. Normally, β -catenin is continuously targeted for ubiquitin-directed proteasomal degradation. However, Wnt binding to Frizzled (and other coreceptors) recruits yet another intracellular protein (*Dishevelled*) that leads to disruption of the degradation-targeting complex. This stabilizes β -catenin, allowing it to translocate to the nucleus and form a transcription complex.

Modular Signaling Proteins, Hubs, and Nodes

The traditional linear view of signaling—that receptor activation triggers an orderly sequence of biochemical intermediates that ultimately leads to changes in gene expression and the desired biological response—is oversimplified. Instead, it is increasingly clear that any initial signal impacts multiple processes, each of which contributes to the final outcome. This is particularly true of signaling pathways that rely on enzymatic activity. For example, specific phosphorylation of any given protein can allow it to associate with a host of other molecules, resulting in (among other effects):

- Enzyme activation (or inactivation)
- Nuclear (or cytoplasmic) localization of transcription factors (see later)
- Transcription factor activation (or inactivation)
- Actin polymerization (or depolymerization)
- Protein degradation (or stabilization)
- Activation of feedback inhibitory (or stimulatory) loops

Adaptor proteins play a key role in organizing intracellular signaling pathways. These proteins function as molecular connectors that physically link different enzymes and promote the assembly of complexes. Adaptors can be integral membrane proteins or cytosolic proteins and typically contain specific domains (e.g., SH2 or SH3) that mediate protein-protein interactions. By influencing which proteins are recruited to signaling complexes, adaptors can determine downstream signaling events.

By analogy with computer networks, the protein-protein complexes can be considered nodes, and the biochemical events feeding into or emanating from these nodes can be thought of as hubs. Signal transduction can therefore be visualized as a kind of networking phenomenon; understanding this higher order complexity is the province of *systems biology*, involving a melding of biology and computer science (i.e., computational biology).

Transcription Factors

Most signal transduction pathways ultimately induce durable effects on cellular function by modulating gene transcription; this occurs through the activation and/or nuclear localization of transcription factors. Some transcription factors drive expression of a relatively limited set of genes or a specific genetic program, while others have widespread effects. Among the transcription factors that regulate cell division are products of several growth-promoting genes, such as *MYC* and *JUN*, and of cell cycle-inhibiting genes, such as *TP53*. Transcription factors often contain modular domains that bind DNA, small molecules such as steroid hormones, and intracellular regulatory proteins. Interactions mediated by these domains can be controlled by posttranslational modifications such as phosphorylation. These changes can result in

translocation from the cytoplasm into the nucleus, modify transcription factor protein half-life, expose specific DNA-binding motifs, or promote binding to components of the RNA polymerase complex to augment transcription factor activity.

- *DNA-binding domains* permit specific binding to short DNA sequences. Whereas some transcription factor binding sites are found in promoters near the location of transcription initiation, other transcription factor binding sites are found in long-range regulatory elements called enhancers. Enhancers loop back to promoters, often in a cell lineage-specific fashion, allowing individual genes to be differentially regulated by the same signal in distinct cell types. The fact that these sites may be distant from one another based on the linear genetic sequence emphasizes the importance of chromatin organization in regulating gene expression.
- *Protein-protein interaction domains* within transcription factors directly or indirectly recruit additional proteins including coactivators, histone-modifying enzymes, and chromatin-remodeling complexes that unwind and/or otherwise expose initiation sites. Most importantly, they recruit RNA polymerase—the large multiprotein enzymatic complex that is responsible for RNA synthesis.

GROWTH FACTORS AND RECEPTORS

Growth factors stimulate the activity of signaling pathways and genes that augment cell survival, growth, and division. Growth factors bind to specific receptors and, ultimately, influence expression of genes that:

- Promote entry into the cell cycle
- Relieve blocks on cell cycle progression (thus promoting replication)
- Prevent apoptosis
- Enhance synthesis of components (nucleic acids, proteins, lipids, carbohydrates) required for cell division

Although growth factors are characteristically thought of as proteins that “just” stimulate cell proliferation and/or survival, it is important to remember that they can also regulate a host of nongrowth activities including migration, differentiation, and synthetic capacity. Some growth factors relevant to tissue repair and regeneration are listed in [Table 1.1](#).

Growth factors regulate cellular proliferation at steady state and in response to injury, when irreversibly damaged cells must be replaced. Uncontrolled proliferation can result when the growth factor activity is dysregulated or when growth factor signaling pathways are altered to become constitutively active. Thus, many growth factor pathway genes are *proto-oncogenes*; by virtue of their proliferative effects, gain-of-function mutations convert them into *oncogenes* that lead to unfettered cell division, one of the hallmarks of cancer (Chapter 7). Although the growth factors described here all involve receptors with intrinsic kinase activity, growth factors can signal through each of the pathways shown in [Fig. 1.13](#).

Epidermal Growth Factor and Transforming Growth Factor- α . Both factors belong to the epidermal growth factor (EGF) family, bind to overlapping sets of receptors, and share many biologic activities. EGF and transforming growth factor- α (TGF- α), which are produced by macrophages and some epithelial cells, are mitogenic for hepatocytes, fibroblasts, and a host of epithelial cell types. The EGF receptor family includes four membrane receptors with intrinsic tyrosine kinase activity; the best characterized receptor is EGFR1, also known as ERBB1, or simply EGFR. EGFR1 mutations and/or amplification frequently occur in cancers, particular those arising in the lung, head and neck, breast, and brain. Another member of the EGFR family, the ERBB2 receptor (also known as *HER-2*), is overexpressed in a subset of breast cancers. Antibodies and small-molecule antagonists that target many of these receptors have proven effective in some cancers.

TABLE 1.1 Growth Factors Involved in Regeneration and Repair

Growth Factor	Sources	Functions
Epidermal growth factor (EGF)	Activated macrophages, salivary glands, keratinocytes, many other cells	Mitogenic for many cell types; stimulates epithelial cell migration; stimulates formation of granulation tissue
Transforming growth factor- α (TGF- α)	Activated macrophages, keratinocytes, many other cells	Stimulates proliferation of hepatocytes and many other epithelial cells
Hepatocyte growth factor (HGF) (scatter factor)	Fibroblasts, stromal cells in the liver, endothelial cells	Enhances proliferation of hepatocytes and other epithelial cells; increases cell motility
Vascular endothelial growth factor (VEGF)	Mesenchymal cells	Stimulates proliferation of endothelial cells; increases vascular permeability
Platelet-derived growth factor (PDGF)	Platelets, macrophages, endothelial cells, smooth muscle cells, keratinocytes	Chemotactic for neutrophils, macrophages, fibroblasts, and smooth muscle cells; activates and stimulates proliferation of fibroblasts, endothelial cells, and other cells; stimulates ECM protein synthesis
Fibroblast growth factors (FGFs) including acidic (FGF-1) and basic (FGF-2)	Macrophages, mast cells, endothelial cells, many other cell types	Chemotactic and mitogenic for fibroblasts; stimulates angiogenesis and ECM protein synthesis
Transforming growth factor- β (TGF- β)	Platelets, T lymphocytes, macrophages, endothelial cells, epithelial cells, smooth muscle cells, fibroblasts	Chemotactic for leukocytes and fibroblasts; stimulates ECM protein synthesis; suppresses acute inflammation
Keratinocyte growth factor (KGF) (i.e., FGF-7)	Fibroblasts	Stimulates keratinocyte migration, proliferation, and differentiation

ECM, Extracellular matrix.

Hepatocyte Growth Factor. Hepatocyte growth factor (HGF; also known as scatter factor) has mitogenic effects on hepatocytes and most epithelium, including the epithelium of the biliary tract, lung, kidney, breast, and skin. HGF acts as a *morphogen* in embryonic development (i.e., it influences the pattern of tissue differentiation), promotes cell migration (hence the designation scatter factor), and enhances hepatocyte survival. HGF is produced by fibroblasts and most mesenchymal cells, endothelial cells, and nonhepatocyte liver cells. It is synthesized as an inactive precursor (pro-HGF) that is proteolytically activated by serine proteases released at sites of injury. The receptor for HGF is MET, which has intrinsic tyrosine kinase activity. MET is frequently overexpressed or mutated in tumors, particularly renal, thyroid papillary carcinomas, and a subset of lung carcinomas. In fact, MET inhibitors are now used routinely to treat lung cancers with MET mutations or overexpression.

Platelet-Derived Growth Factor. Platelet-derived growth factor (PDGF) is a family of several closely related proteins, each consisting of two chains (designated by pairs of letters). Three isoforms of PDGF (AA, AB, and BB) are directly biologically active; two others, PDGF-CC and PDGF-DD, must be activated by proteolytic cleavage. PDGF proteins are stored in cytoplasmic granules and released by activated platelets. Although originally isolated from platelets (hence the name), PDGFs are produced by many cells including activated macrophages, endothelium, smooth muscle cells, and tumors. All PDGF isoforms exert their effects by binding to two cell-surface receptors (PDGFR α and β), both of which have intrinsic tyrosine kinase activity. PDGF induces fibroblast, endothelial, and smooth muscle cell proliferation and is also chemotactic for these cells (and inflammatory cells), thereby promoting their recruitment to sites of inflammation and tissue injury.

Vascular Endothelial Growth Factor. Vascular endothelial growth factor (VEGF) is a family of homodimeric proteins: VEGF-A, VEGF-B, VEGF-C, VEGF-D, and placental growth factor (PlGF). VEGF-A is generally referred to simply as VEGF; it is the major *angiogenic* factor (inducing blood vessel development) after injury and in tumors. In comparison, VEGF-B and PlGF are involved in embryonic vessel development, and VEGF-C and VEGF-D stimulate both angiogenesis and lymphatic development (*lymphangiogenesis*). Separate from their

roles in angiogenesis, VEGFs are also involved in the maintenance of normal endothelium, with highest expression in epithelial cells adjacent to fenestrated epithelium (e.g., kidney podocytes, retinal pigment epithelium, and choroid plexus). VEGF induces all the activities necessary for angiogenesis, including endothelial cell migration and proliferation (capillary sprouting), and promotes formation of vascular lumina. VEGF also affects vascular dilation and increases vascular permeability (VEGF was originally called *vascular permeability factor* to reflect that activity). As might be anticipated, hypoxia is the most important inducer of VEGF production, through pathways that involve activation of the transcription factor *hypoxia-inducible factor 1* (HIF-1). Other VEGF inducers—produced at sites of inflammation or wound healing—include PDGF and TGF- α .

VEGFs bind to a family of tyrosine kinase receptors (VEGFR-1, VEGFR-2, and VEGFR-3); VEGFR-2 is highly expressed in endothelium and is the most important for angiogenesis. Antibodies against VEGF are approved for the treatment of tumors such as renal and colon cancers that require angiogenesis for their spread and growth. Anti-VEGF therapies have had success in ophthalmic disorders including “wet” age-related macular degeneration (a disorder of inappropriate angiogenesis and vascular permeability that causes adult-onset blindness), angiogenesis associated with retinopathy of prematurity, and vascular leakage that leads to diabetic macular edema. Finally, increased levels of soluble VEGFR-1 (also known as s-FLT-1) in pregnant women may cause preeclampsia (hypertension and proteinuria) by sequestering free VEGF, which is required for maintenance of normal endothelial function.

Fibroblast Growth Factor. Fibroblast growth factor (FGF) refers to a family of growth factors with more than 20 members. Acidic FGF (aFGF) (also known as FGF-1) and basic FGF (bFGF) (also known as FGF-2) are the best characterized; FGF-7 is also referred to as keratinocyte growth factor (KGF). Released FGFs associate with heparan sulfate in the ECM, which serves as a reservoir for inactive factors that can be subsequently released by proteolysis (e.g., at sites of wound healing). FGFs signal through four tyrosine kinase receptors (FGFR1 through FGFR4) to promote wound healing and development; bFGF also has all the activities necessary for angiogenesis.

Transforming Growth Factor- β . TGF- β has three isoforms (TGF- β 1, TGF- β 2, TGF- β 3) that belong to a larger family of about 30 members including bone morphogenetic proteins (BMPs), activins, inhibins, and müllerian inhibiting substance. TGF- β 1 has the most widespread distribution and is commonly referred to simply as TGF- β ; it is a homodimeric protein produced by multiple cell types including platelets, endothelium, epithelial cells, and inflammatory cells. TGF- β is secreted as a precursor that requires proteolysis to yield the biologically active protein. There are two TGF- β receptors (types I and II) both with serine/threonine kinase activity that induce the phosphorylation of several related downstream transcription factors called *Smads*. Phosphorylated Smads form heterodimers, allowing nuclear translocation and association with other DNA-binding proteins to activate or inhibit gene transcription. TGF- β signaling has multiple—and often opposing—effects, depending on the tissue type and other concurrent signals. However, two of the principal effects of TGF- β are the promotion of fibrosis (scar formation) and inhibition of the inflammation that accompanies wound healing.

- TGF- β stimulates the production of collagen, fibronectin, and proteoglycans and inhibits collagen degradation by decreasing matrix metalloproteinase (MMP) activity and increasing the activity of tissue inhibitors of metalloproteinases (TIMPs) (discussed later). TGF- β is involved not only in healing (scar formation) after injury but also drives fibrosis in lung, liver, intestines, and kidneys in the setting of chronic inflammation that impairs tissue function.
- TGF- β is also an antiinflammatory cytokine that serves to limit and terminate inflammatory responses. It does this by inhibiting lymphocyte proliferation and the activity of other leukocytes. Animals lacking TGF- β have widespread and persistent inflammation.

EXTRACELLULAR MATRIX

The ECM is a protein network that constitutes a significant proportion of any tissue. Cell interactions with the ECM are critical for development, healing, and maintenance of normal tissue architecture (Fig. 1.14). Much more than a simple “space filler” around cells, the ECM functions as a:

- *Mechanical support* for cell anchorage, cell migration, and maintenance of cell polarity.
- *Regulator of cell proliferation* by binding and displaying growth factors and by signaling via cellular integrin family receptors. The ECM provides a depot for latent growth factors that can be activated within foci of injury or inflammation.
- *Scaffold for tissue renewal.* Because maintenance of normal tissue structure requires a basement membrane or stromal scaffolds, integrity of the basement membrane or the stroma of parenchymal cells is critical for organized tissue regeneration. Thus, ECM disruption prevents effective tissue regeneration and repair.
- *Foundation for establishment of tissue microenvironments.* Basement membrane acts as a boundary between epithelium and underlying connective tissue but often does more than just provide structural support; for example, in the kidney, it forms part of the filtration apparatus.

The ECM is constantly being remodeled; its synthesis and degradation accompany morphogenesis, tissue regeneration and repair, chronic fibrosis, and tumor invasion and metastasis.

The ECM occurs in two basic forms: interstitial matrix and basement membrane (Fig. 1.15).

- *Interstitial matrix.* Interstitial matrix occupies the spaces between stromal cells within connective tissue and between parenchymal

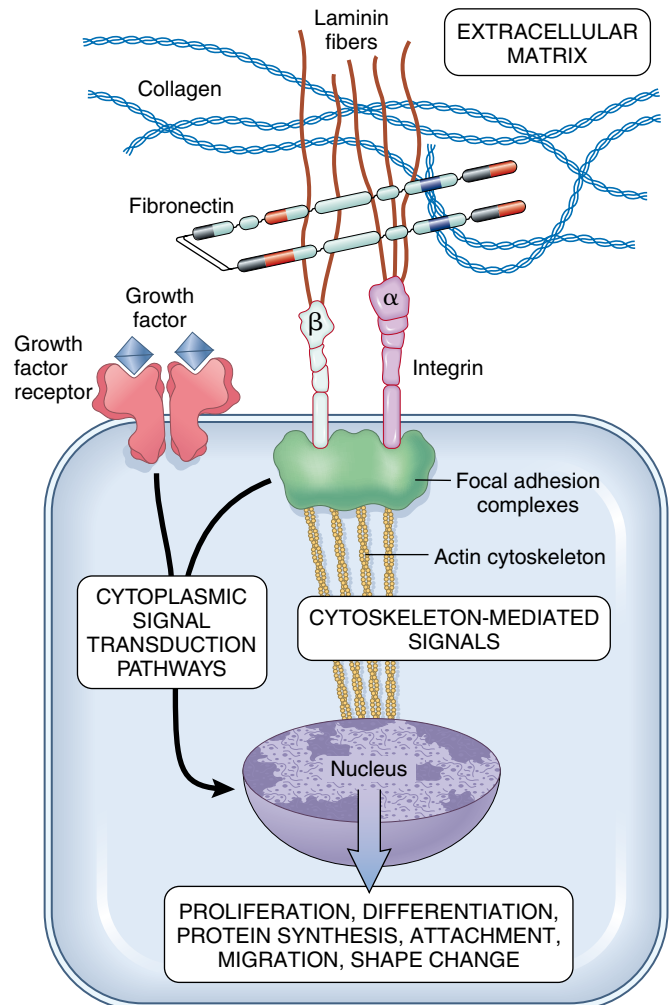


Fig. 1.14 Interactions of extracellular matrix (ECM) and growth factors to mediate cell signaling. Cell-surface integrins interact with the cytoskeleton at focal adhesion complexes (protein aggregates that include vinculin, α -actinin, and talin; see Fig. 1.18C). This initiates the production of intracellular messengers or directly transduces signals to the nucleus. Cell-surface receptors for growth factors can activate signal transduction pathways that overlap with those mediated through integrins. Signals from both ECM interactions and growth factors are integrated by the cells to produce specific responses, including changes in proliferation, locomotion, or differentiation.

epithelium and the underlying supportive vascular and smooth muscle structures in some organs. Interstitial matrix is synthesized by mesenchymal cells (e.g., fibroblasts), forming a three-dimensional, relatively amorphous, semi-fluid gel. In some tissues, such as the gastrointestinal tract, urinary bladder, and periarterial soft tissues, fluid within the matrix cushions tissue compression associated with peristalsis, urination, and pulsatile arterial blood flow. The major nonfluid constituents of the interstitial matrix are fibrillar and nonfibrillar collagens, as well as *fibronectin*, *elastin*, *proteoglycans*, *hyaluronate*, and other constituents (see later).

- *Basement membrane.* Interstitial matrix within connective tissues becomes highly organized around epithelial cells, endothelial cells, and smooth muscle cells, where it forms basement membranes, specialized surfaces for cell growth. Basement membrane compo-

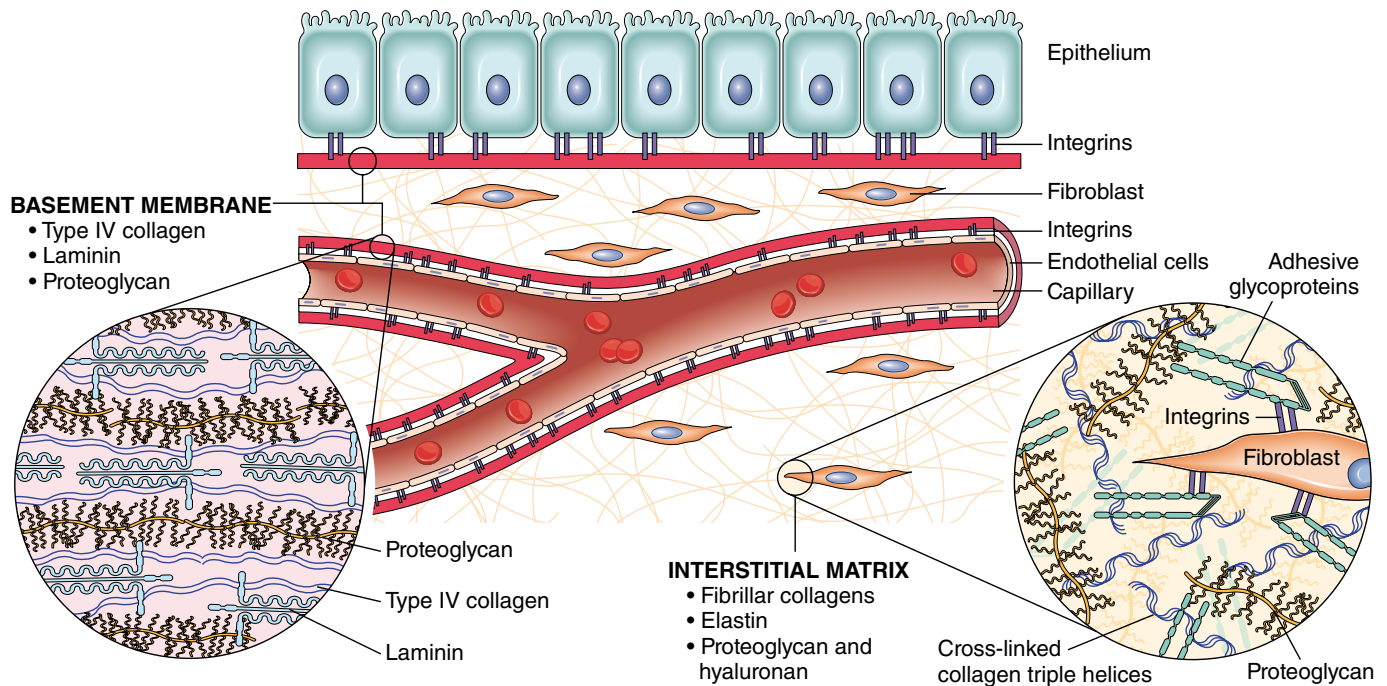


Fig. 1.15 Main components of the extracellular matrix (ECM) including collagens, proteoglycans, and adhesive glycoproteins. Both epithelial and mesenchymal cells (e.g., fibroblasts) interact with ECM via integrins. Basement membranes and interstitial ECM have different architecture and general composition, although certain components are present in both. For the sake of clarity, many ECM components (e.g., elastin, fibrillin, hyaluronan, and syndecan) are not shown.

nents, which are synthesized by the overlying epithelium and underlying mesenchymal cells, form a flat lamellar mesh (although labeled as a membrane, it is quite porous). The major constituents are nonfibrillar *type IV collagen* and *laminin*.

Components of the Extracellular Matrix

ECM components fall into three families (Fig. 1.16):

- *Fibrous structural proteins* such as collagens and elastin that confer tensile strength and recoil
- *Water-hydrated gels* such as proteoglycans and hyaluronan that permit compressive resistance and lubrication
- *Adhesive glycoproteins* that connect ECM elements to one another and to cells

Collagens. Collagens are composed of three separate polypeptide chains braided into a ropelike triple helix (Fig. 1.17). About 30 collagen types have been identified, some of which are unique to specific cells and tissues.

- *Fibrillar collagens:* Some collagen types (e.g., types I, II, III, and V) form linear fibrils stabilized by interchain hydrogen bonding; such fibrillar collagens comprise a major proportion of the connective tissue in bone, tendon, cartilage, blood vessels, and skin, as well as in healing wounds and scars. The tensile strength of the fibrillar collagens derives from lateral cross-linking of the triple helices via covalent bonds whose formation requires lysine hydroxylation. The responsible enzyme, *lysyl hydroxylase*, is dependent on vitamin C, explaining why children with ascorbate deficiency have skeletal deformities and why individuals of any age with vitamin C deficiency (scurvy, Chapter 9) heal poorly and bleed easily. Genetic defects, including collagen and lysyl

hydroxylase mutations, cause diseases such as *osteogenesis imperfecta* and certain forms of *Ehlers-Danlos syndrome* (Chapter 5).

- *Nonfibrillar collagens* (e.g., type IV collagen) contribute to the structure of planar basement membranes; help regulate collagen fibril diameters and collagen-collagen interactions via so-called fibril-associated collagen with interrupted triple helices (FACITs) (e.g., type IX collagen in cartilage); or provide anchoring fibrils that maintain structure of stratified squamous epithelium (e.g., type VII collagen. Mutations lead to blistering skin diseases).

Elastin. The ability of tissues to elastically recoil and return to a baseline structure after physical stress is conferred by elastin (see Fig. 1.16A). Elasticity is especially important in cardiac valves and large blood vessels, which need to accommodate recurrent pulsatile flow, as well as in the uterus, skin, and ligaments. Morphologically, elastic fibers consist of a central core of elastin with an associated meshlike network of *fibrillin* glycoprotein. The latter relationship partially explains why fibrillin synthetic defects lead to skeletal abnormalities and weakened aortic walls, as in patients with Marfan syndrome; fibrillin also controls the availability of free TGF- β , and this function plays a role in pathogenesis of Marfan syndrome (Chapter 5).

Proteoglycans and Hyaluronan. Proteoglycans form highly hydrated compressible gels that confer resistance to compressive forces; in joint cartilage, proteoglycans also provide a layer of lubrication between adjacent bony surfaces. Proteoglycans consist of long polysaccharides, called *glycosaminoglycans* (e.g., keratan sulfate and chondroitin sulfate) attached to a core protein; these are then linked to a long hyaluronic acid polymer called *hyaluronan*, in a manner reminiscent of the bristles on a round brush. The highly negatively charged nature of the densely

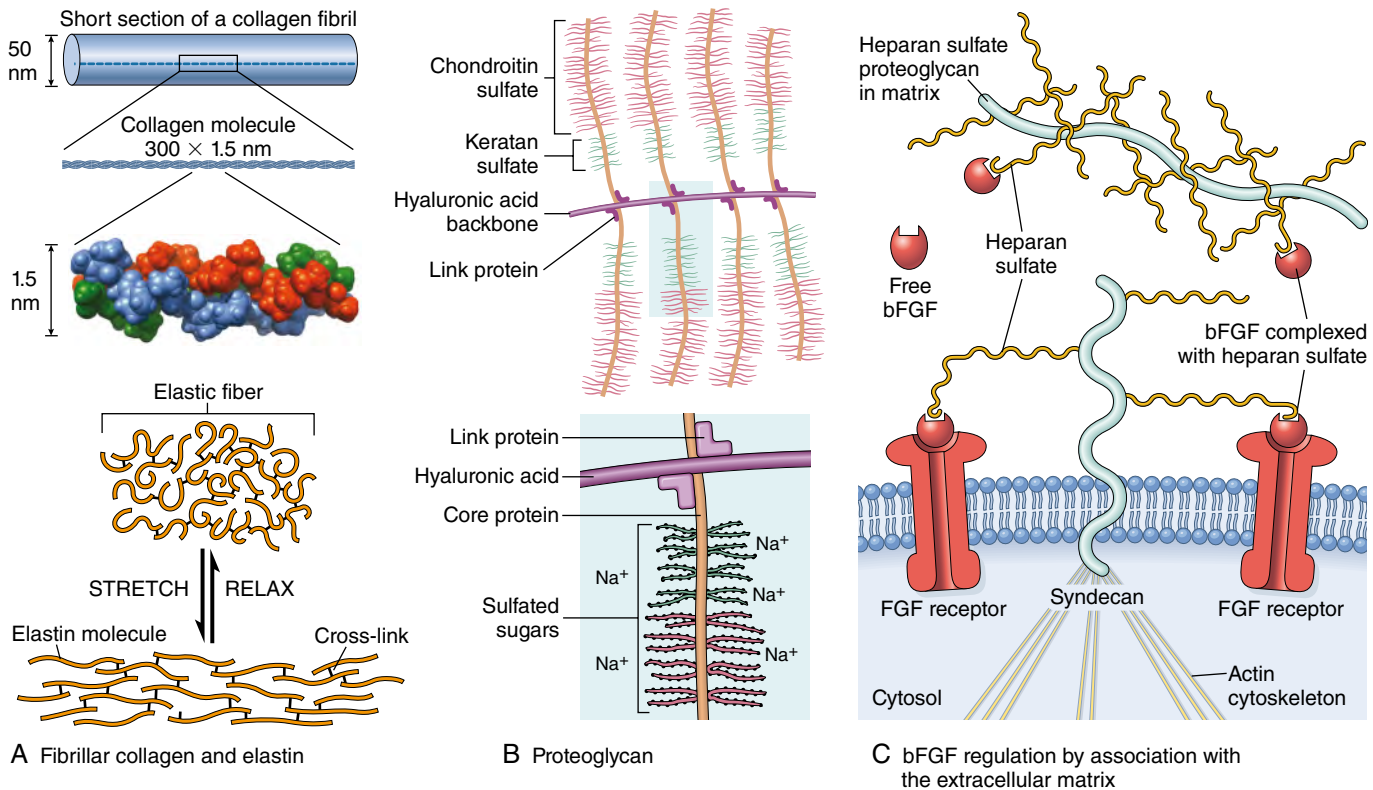


Fig. 1.16 Extracellular matrix (ECM) components. (A) Fibrillar collagen and elastic tissue structures. Due to rodlike fibril stacking and extensive lateral cross-linking, collagen fibers have marked tensile strength but lack elasticity. Elastin is also cross-linked but differs in having large hydrophobic segments that form a dense globular configuration at rest. As stretch is exerted, the hydrophobic domains are pulled open, but the cross-links keep the tissue intact; release of the stretch tension allows the hydrophobic domains of the proteins to recoil. (B) Proteoglycan structure. The highly negatively charged sulfated sugars on the proteoglycan “bristles” attract sodium and water to generate a viscous but compressible matrix. (C) Regulation of basic fibroblast growth factor (bFGF; also known as FGF-2) activity by ECM and cellular proteoglycans. Heparan sulfate binds bFGF secreted in the ECM. Syndecan is a cell-surface proteoglycan with a transmembrane core protein and extracellular glycosaminoglycan side chains, such as heparan sulfate, that can bind bFGF and a cytoplasmic tail that interacts with the intracellular actin cytoskeleton. Syndecan side chains bind bFGF released from damaged ECM, thus facilitating bFGF interaction with cell-surface receptors. *FGF*, Fibroblast growth factor.

packed sulfated sugars attracts cations (mostly sodium) and with them, abundant osmotically attracted water—producing a viscous, gel-like matrix. Besides providing compressibility to tissues, proteoglycans also serve as reservoirs for growth factors secreted into the ECM (e.g., FGF and HGF). Some proteoglycans are integral cell membrane proteins that have roles in cell proliferation, migration, and adhesion (e.g., by binding and concentrating growth factors and chemokines) (see Fig. 1.16C).

Adhesive Glycoproteins and Adhesion Receptors. Adhesive glycoproteins and adhesion receptors are structurally diverse molecules variously involved in cell-cell, cell-ECM, and ECM-ECM interactions (Fig. 1.18). Prototypical adhesive glycoproteins include *fibronectin* (a major component of the interstitial ECM) and *laminin* (a major constituent of basement membrane). *Integrins* are representative of the adhesion receptors, also known as cell adhesion molecules (CAMs); the CAMs also include immunoglobulin family members, cadherins, and selectins.

- *Fibronectin* is a large (450-kDa), disulfide-linked heterodimer that exists in tissue and plasma forms; it is synthesized by a variety of cells including fibroblasts, monocytes, and endothelium. Fibronectin has specific domains that can bind to distinct ECM components (e.g., collagen, fibrin, heparin, and proteoglycans), as well as attach

to cell integrins (see Fig. 1.18A). In healing wounds, tissue and plasma fibronectin provide the scaffolding for subsequent ECM deposition, angiogenesis, and reepithelialization.

- *Laminin* is the most abundant glycoprotein in basement membranes. It is an 820-kDa cross-shaped heterotrimer that connects cells to underlying ECM components such as type IV collagen and heparan sulfate (see Fig. 1.18B). Besides mediating cell attachment to basement membrane, laminin can also modulate cell proliferation, differentiation, and motility.
- *Integrins* are a large family of transmembrane heterodimeric glycoproteins composed of α - and β -subunits; these allow cells to attach to ECM constituents such as laminin and fibronectin, thus functionally and structurally linking the intracellular cytoskeleton with the outside world. Integrins also facilitate cell-cell adhesive interactions. On leukocytes, they mediate the firm adhesion to and migration across endothelium and epithelium at sites of inflammation (Chapter 3), and they also play a critical role in platelet aggregation (Chapter 4). Integrins attach to ECM components via a tripeptide arginine-glycine-aspartic acid motif (abbreviated RGD). Besides providing focal attachment to underlying substrates, binding through the integrin receptors can trigger signaling cascades that regulate cell locomotion, proliferation, shape, and differentiation (see Fig. 1.18C).

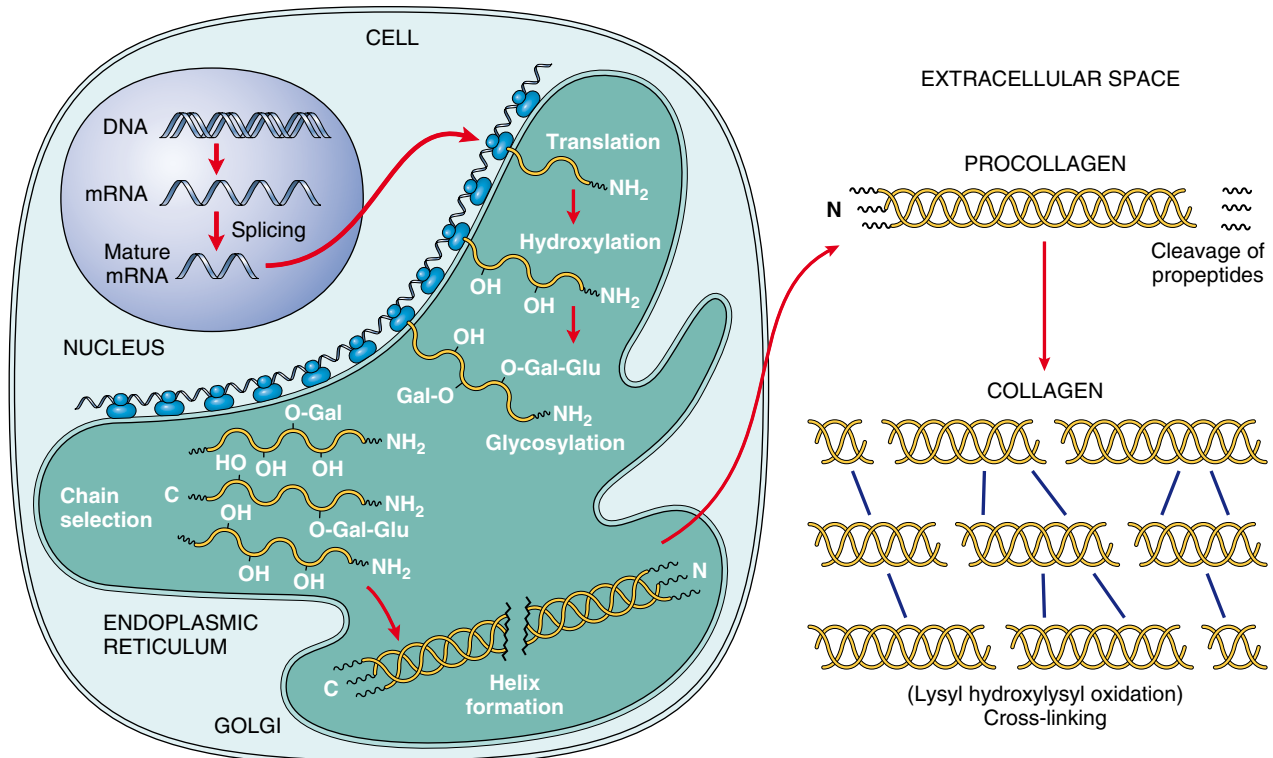


Fig. 1.17 Collagen biosynthetic pathway. Some collagen types are heterotrimers (types I, V, and XI), while others are homotrimers (types II and III). The α -chains that make up fibrillar collagen molecules are synthesized as precursor pro- α -chains, with large globular polypeptide regions flanking the central triple-helical domain. After proline and lysine hydroxylation and lysine glycosylation within the endoplasmic reticulum, three procollagen chains align to form a triple helix. For all the fibrillary collagens, the carboxyl end of the propeptide is completely removed by endoprotease activity after secretion, and the resulting triple-helical rodlike domains polymerize in a staggered fashion to form fibrils. The N-terminus propeptide is variably processed depending on the collagen chain. For collagen types I and II, the N-propeptide processing is complete, while for collagens V and XI, a large portion of the N-propeptides remain attached; such “incomplete” processing can regulate fibril size. After secretion, collagen achieves lateral stability through cross-linking involving lysyl oxidase and the previously hydroxylated residues. Defects in primary sequence, procollagen endopeptidase processing, hydroxylation, or cross-linking can all lead to connective tissue laxity. The specific structures (e.g., blood vessels, skin, bone, ligaments) affected by such disorders are predictable based on the collagen that predominates in that tissue. *mRNA*, Messenger RNA.

MAINTAINING CELL POPULATIONS

Proliferation and the Cell Cycle

Cell proliferation is fundamental to organism development, to maintenance of steady-state tissue homeostasis, and to replacement of dead or damaged cells. The key elements of cellular proliferation are accurate DNA replication, coordinated synthesis of other cellular components, and equal apportionment of DNA and organelles to daughter cells through the processes of mitosis and cytokinesis.

The sequence of events that results in cell proliferation is called the *cell cycle*, which consists of G_1 (gap 1), S (DNA synthesis), G_2 (gap 2), and M (mitotic) phases. Quiescent cells that are not actively cycling are in the G_0 (gap 0) state (Fig. 1.19). Cells can enter G_1 either from the G_0 quiescent cell pool or after completing a round of mitosis. Each stage requires completion of the previous step, as well as activation of necessary factors (see later). Infidelity of DNA replication or cofactor deficiency results in arrest at various transition points.

The cell cycle is regulated by activators and inhibitors. Cell cycle progression is chaperoned by proteins called *cyclins*—named for the cyclic nature of their production and degradation—and cyclin-associated enzymes called *cyclin-dependent kinases* (CDKs) (Fig. 1.20).

Constitutively synthesized CDKs acquire kinase activity—that is, the ability to phosphorylate protein substrates—by forming complexes with the relevant cyclins. Transiently increased synthesis of a particular cyclin thus leads to increased kinase activity of the appropriate CDK binding partner; as the CDK completes its round of phosphorylation, the associated cyclin is degraded and CDK activity abates. Consequently, as cyclin levels rise and fall, activity of associated CDKs likewise waxes and wanes.

More than 15 cyclins have been identified; cyclins D, E, A, and B appear sequentially during the cell cycle and bind to one or more CDKs. The cell cycle thus resembles a relay race in which each leg is regulated by a distinct set of cyclins: as one collection of cyclins leaves the track, the next set takes over.

Surveillance mechanisms primed to sense DNA or chromosomal damage are embedded within the cell cycle. These quality control *checkpoints* ensure that cells with genetic imperfections do not complete replication. Thus, the G_1 -S checkpoint monitors DNA integrity before irreversibly committing cellular resources to DNA replication. Later in the cell cycle, the G_2 -M restriction point ensures that there has been accurate genetic replication before the cell divides. When cells do detect DNA irregularities, checkpoint activation delays cell cycle

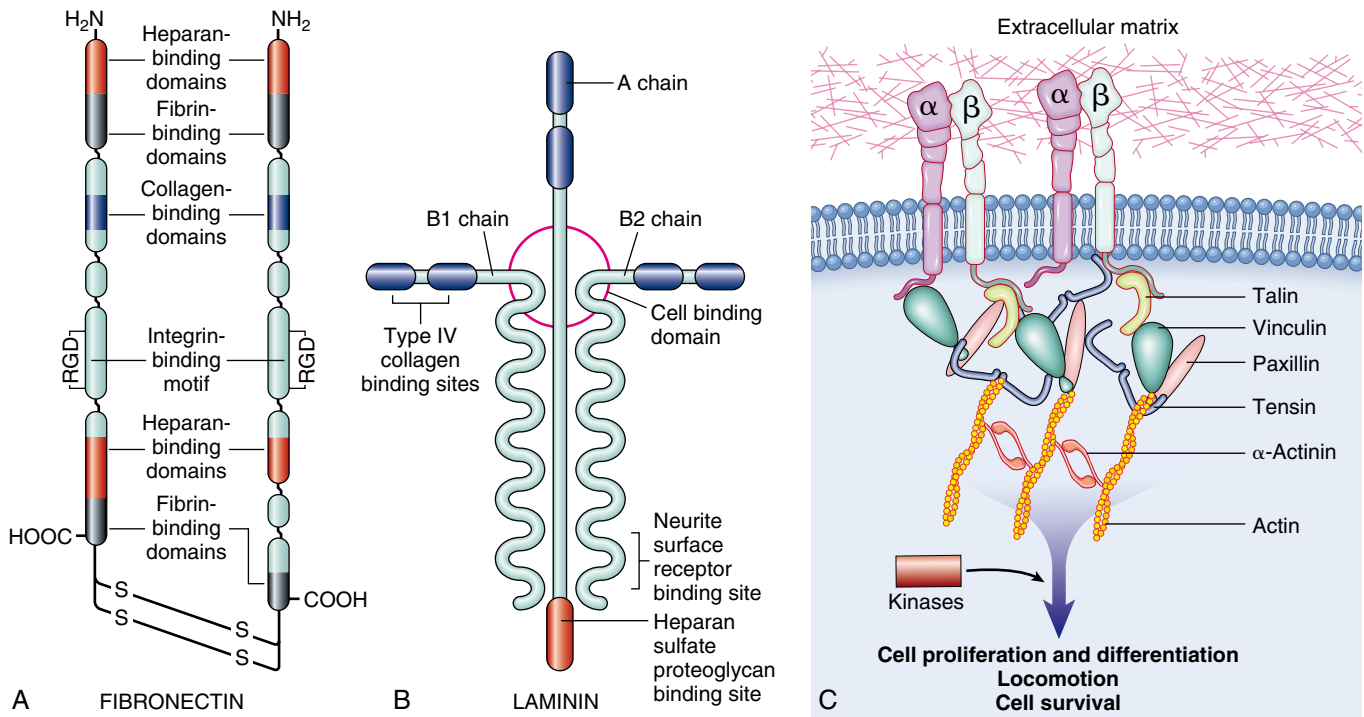


Fig. 1.18 Cell and extracellular matrix (ECM) interactions: adhesive glycoproteins and integrin signaling. (A) Fibronectin consists of a disulfide-linked dimer, with several distinct domains that allow binding to ECM and to integrins, the latter through arginine-glycine-aspartic acid (*RGD*) motifs. (B) The cross-shaped laminin molecule is one of the major components of basement membranes; its multidomain structure allows interactions between type IV collagen, other ECM components, and cell-surface receptors. (C) Integrins and integrin-mediated signaling events at focal adhesion complexes. Each α - β heterodimeric integrin receptor is a transmembrane dimer that links ECM and the intracellular cytoskeleton. Focal adhesion complexes include linking molecules (e.g., vinculin and talin) that can recruit and activate kinases and, ultimately, trigger downstream signaling cascades.

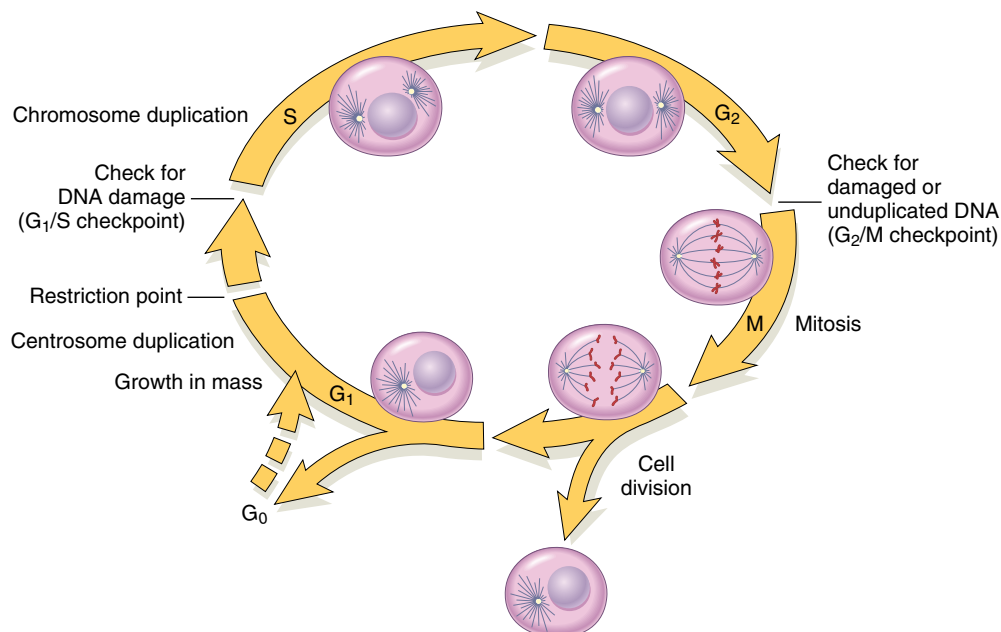


Fig. 1.19 Cell cycle landmarks. The figure shows the cell cycle phases (G_0 , G_1 , G_2 , S , and M), the location of the G_1 restriction point, and the G_1/S and G_2/M cell cycle checkpoints. G_1 restriction point refers to the stage in G_1 where the cell is committed to advance further into the cell cycle without requiring any more of the growth signal that initiated cell division. Cells from labile tissues such as the epidermis and the gastrointestinal tract may cycle continuously; stable cells such as hepatocytes are quiescent but can enter the cell cycle; permanent cells such as neurons and cardiac myocytes have lost the capacity to proliferate. (Modified from Pollard TD, Earnshaw WC. *Cell Biology*. Saunders; 2002.)

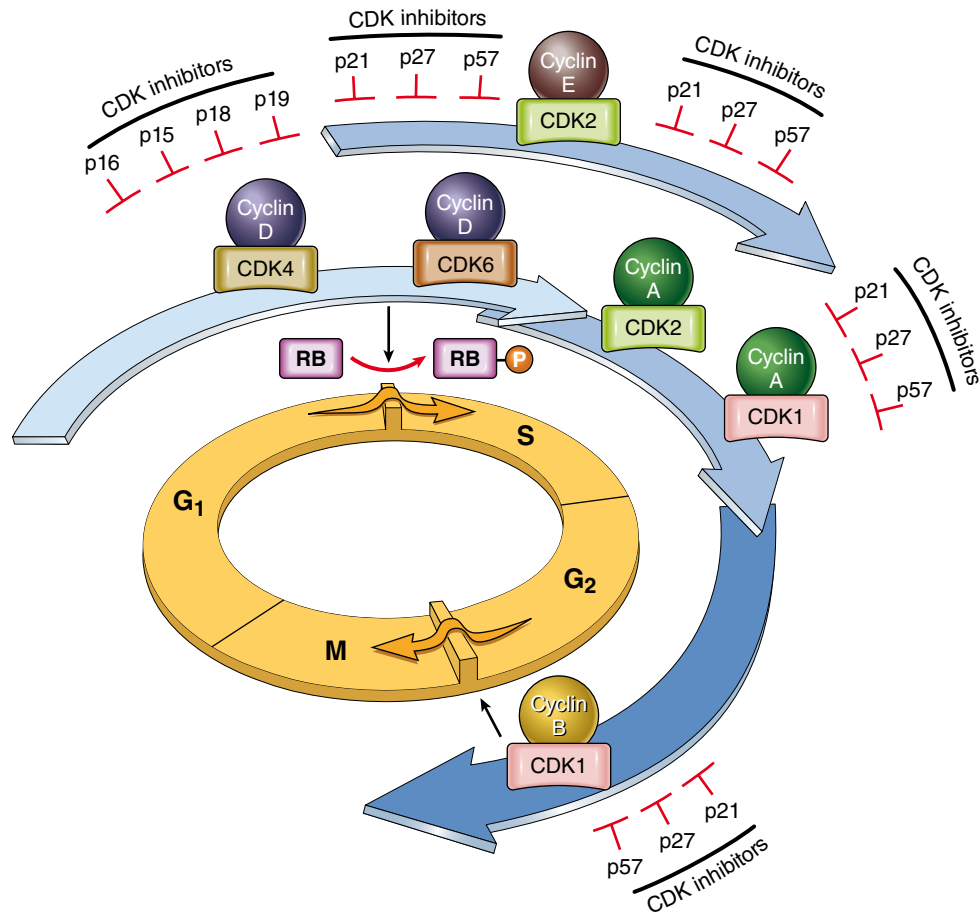


Fig. 1.20 Role of cyclins, cyclin-dependent kinases (CDKs), and CDK inhibitors (CKIs) in regulating the cell cycle. Shaded arrows represent the phases of the cell cycle during which specific cyclin-CDK complexes are active. As illustrated, cyclin D-CDK4, cyclin D-CDK6, and cyclin E-CDK2 regulate the G₁-to-S transition by phosphorylating the Rb protein (pRb). Cyclin A-CDK2 and cyclin A-CDK1 are active in the S phase. Cyclin B-CDK1 is essential for the G₂-to-M transition. Two families of CDK inhibitors can block activity of CDKs and progression through the cell cycle. The so-called INK4 inhibitors, composed of p15, p16, p18, and p19, act on cyclin D-CDK4 and cyclin D-CDK6. The other family of three inhibitors, p21, p27, and p57, can inhibit all CDKs.

progression and triggers DNA repair mechanisms. If the genetic derangement is too severe to be repaired, cells either undergo apoptosis or enter a nonreplicative state called *senescence*—primarily through p53-dependent mechanisms (see later).

Enforcing the cell cycle checkpoints is the job of *CDK inhibitors* (CKIs); they accomplish this by modulating CDK-cyclin complex activity. There are several different CKIs.

- One family of CKIs—composed of three proteins called p21 (CDKN1A), p27 (CDKN1B), and p57 (CDKN1C)—broadly inhibits multiple CDKs.
- Another family of CKIs has selective effects on cyclin CDK4 and cyclin CDK6; these proteins are called p15 (CDKN2B), p16 (CDKN2A), p18 (CDKN2C), and p19 (CDKN2D).
- Defective CKI checkpoint proteins allow cells with damaged DNA to divide, resulting in mutated daughter cells at risk for malignant transformation.

An equally important aspect of cell growth and division is the biosynthesis of the membranes, cytosolic proteins, and organelles necessary to make two daughter cells. Thus, as growth factor receptor signaling stimulates cell cycle progression, it also activates events that promote the metabolic changes that support growth. Chief among these is the switch to aerobic glycolysis (with the counter-intuitive

reduction in oxidative phosphorylation), also called the *Warburg effect*. These alterations in cell metabolism are an important element in cancer cell growth and are discussed in greater detail in Chapter 7.

Stem Cells

Stem cells have the dual property of being able to self-renew and to give rise to differentiated cells and tissues. During development, *totipotent stem cells* give rise to the full range of differentiated tissues; in the mature organism, *adult stem cells* only have the capacity to replace damaged cells and maintain cell populations within the tissues where they reside. There are also populations of stem cells between these extremes with varying capacities to differentiate into multiple (but limited) cell lineages. Thus, depending on the source and stage of development, there are limits on the cell types that a “stem cell” can generate.

In normal tissues—without healing, degeneration, or neoplasia—there is a homeostatic equilibrium between replication, self-renewal, and differentiation of stem cells and death of mature, fully differentiated cells (Fig. 1.21). The dynamic relationship between stem cells and terminally differentiated parenchyma is particularly evident in the continuously dividing epithelium of the skin; stem cells at the basal layer of the epithelium divide and their daughter cells progressively

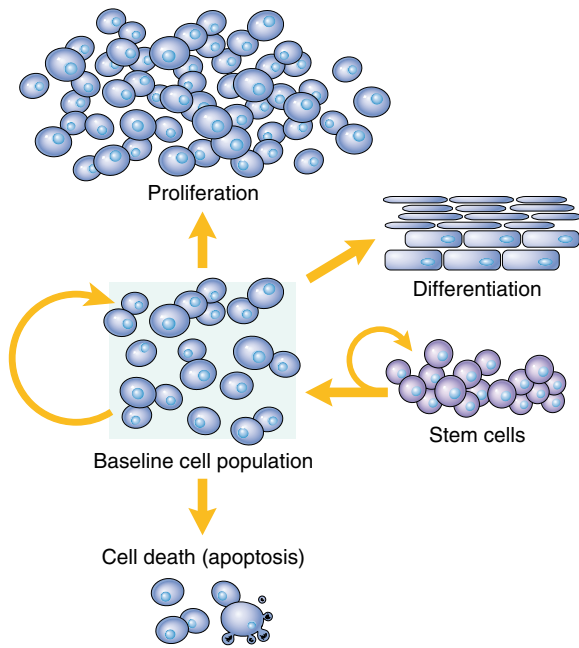


Fig. 1.21 Mechanisms regulating cell populations. Cell numbers can be altered by increased or decreased rates of stem cell input, cell death resulting from apoptosis, or changes in the rates of proliferation or differentiation. (Modified from McCarthy NJ, Smith CA, Williams GT. Apoptosis in the development of the immune system: growth factors, clonal selection and bcl-2. *Cancer Metastasis Rev.* 1992;11(2):157–178.)

differentiate as they migrate to the upper layers of the epithelium before dying and being shed.

Under conditions of homeostasis, stem cells are maintained by self-renewal, which can involve two types of cell division:

- *Asymmetric division* refers to cell replication in which one daughter cell enters a differentiation pathway and gives rise to mature cells, while the other remains undifferentiated and retains its self-renewal capacity.
- *Symmetric division* occurs when both daughter cells retain self-renewal capacity. Such replication occurs early in embryogenesis—when

stem cell populations are expanding—and under stress conditions, as in bone marrow repopulation after ablative chemotherapy.

Although some researchers separate stem cells into multiple different subsets, there are fundamentally only two varieties.

- *Embryonic stem (ES) cells* are the most undifferentiated. They are present in the inner cell mass of the blastocyst, have virtually limitless cell renewal capacity, and can give rise to every cell in the body; they are thus said to be *totipotent* (Fig. 1.22). While ES cells can be maintained for extended periods without differentiating, they can be induced under appropriate culture conditions to form specialized cells of all three germ cell layers.
- *Tissue stem cells* (also called *adult stem cells*) are found in intimate association with the differentiated cells of a given tissue. They are normally protected within specialized tissue microenvironments called *stem cell niches*. Such niches have been demonstrated in many organs, most notably the bone marrow, where hematopoietic stem cells characteristically congregate in perivascular niches, and the intestines, where epithelial stem cells are confined to the crypts. Other stem cell niches include the bulge region of hair follicles, the limbus of the cornea, and the subventricular zone in the brain. Soluble factors and other cells within the niches regulate the balance between stem cell quiescence, expansion, and differentiation (Fig. 1.23).

Adult stem cells have a limited repertoire (*lineage potential*) of differentiated cells that they can generate. Thus, although adult stem cells can maintain tissues with high (e.g., skin, gastrointestinal tract, and bone marrow) or low (e.g., endothelial) cell turnover, the adult stem cells in any given tissue can usually produce only those cells normally found within that tissue.

Hematopoietic stem cells continuously replenish all the cellular elements of the blood as they exit the circulation, senesce, or are otherwise consumed. They can be isolated directly from bone marrow, as well as from the peripheral blood after administration of certain colony-stimulating factors that induce their release from bone marrow niches. Although they are rare, hematopoietic stem cells can be purified to virtual purity based on cell-surface markers. Clinically, these stem cells can be used to repopulate marrows depleted after chemotherapy (e.g.,

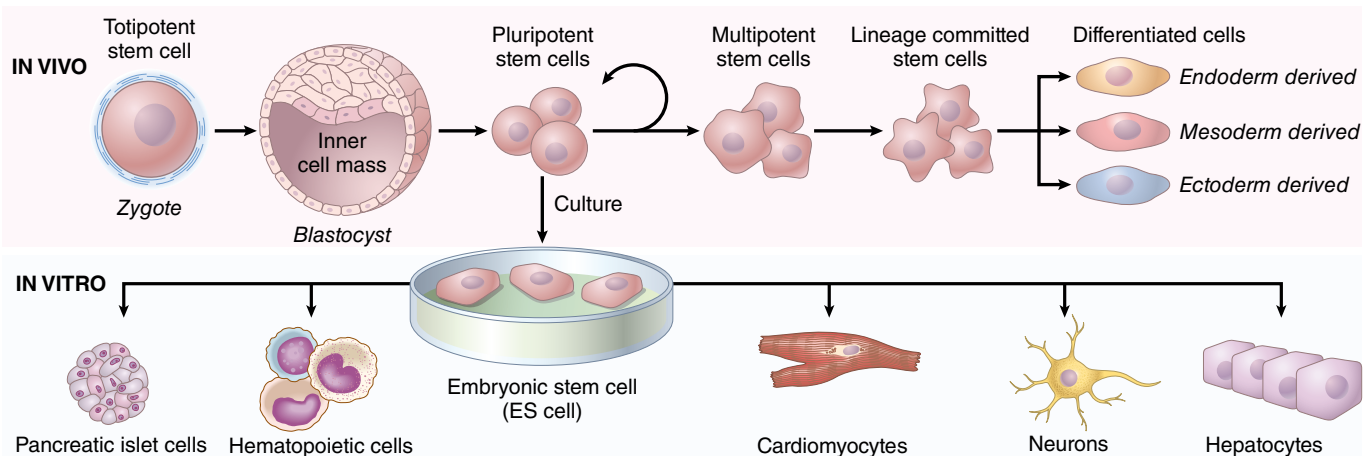


Fig. 1.22 Embryonic stem (ES) cells. The zygote, formed by the union of sperm and egg, divides to form blastocysts, and the inner cell mass of the blastocyst generates the embryo. The pluripotent cells of the inner cell mass, known as ES cells, can be induced to differentiate into cells of multiple lineages. In the embryo, pluripotent stem cells can asymmetrically divide to yield a residual stable pool of ES cells in addition to generating populations that have progressively more restricted developmental capacity, eventually generating stem cells that are committed to just specific lineages. ES cells can be cultured in vitro and induced to differentiate into cells characteristic of all three germ layers.

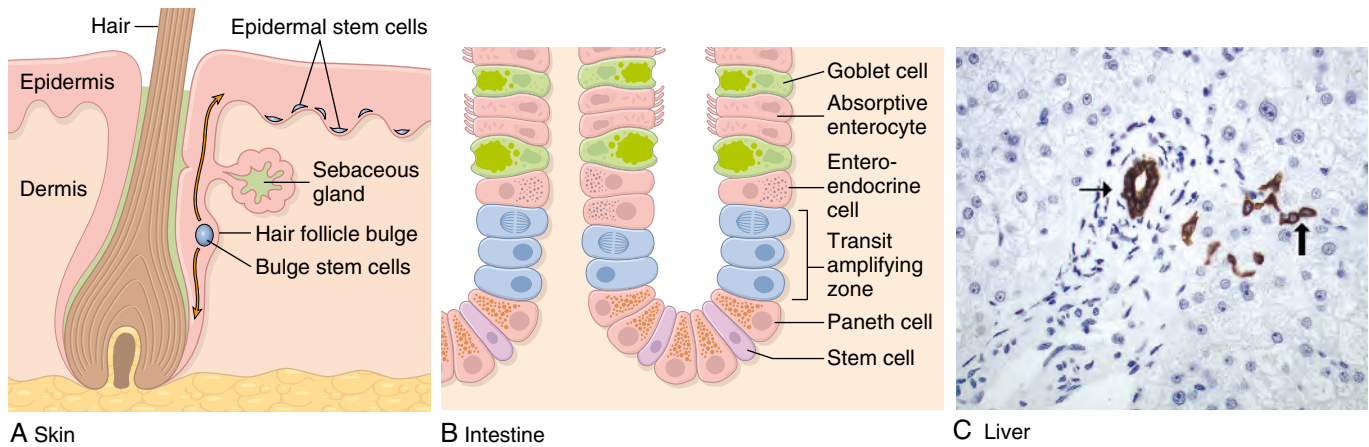


Fig. 1.23 Stem cell niches in various tissues. (A) Skin stem cells are located in the bulge area of the hair follicle, in sebaceous glands, and in the lower layer of the epidermis. (B) Small intestinal crypt base columnar (CBC) stem cells are located at the base of the crypt interspersed between Paneth cells. (C) Liver stem cells (oval cells) are located in the canals of Hering (*thick arrow*), structures that connect bile ductules (*thin arrow*) to parenchymal hepatocytes. Bile duct cells and canals of Hering are highlighted here by an immunohistochemical stain for cytokeratin 7. (C, Courtesy Tania Roskams, MD, University of Leuven, Belgium.)

for leukemia) or to provide normal precursors to correct various blood cell defects (e.g., sickle cell disease; see Chapter 14).

Besides hematopoietic stem cells, the bone marrow (and, notably, other tissues such as fat) also contains a population of *mesenchymal stem cells*, multipotent cells that can differentiate into a variety of stromal cells including chondrocytes (cartilage), osteocytes (bone), adipocytes (fat), and myocytes (muscle). Because these cells can be massively expanded and can generate a locally immunosuppressive microenvironment (thus potentially evading rejection), they represent a potential means of manufacturing stromal cellular scaffoldings for tissue regeneration.

Regenerative Medicine

The burgeoning field of regenerative medicine has been made possible by the ability to identify, isolate, expand, and transplant stem cells. Theoretically, the differentiated progeny of ES or adult stem cells can be used to repopulate damaged tissues or even to construct entire replacement organs. There is, therefore, considerable interest in therapeutic opportunities for restoring tissues that have low intrinsic regenerative capacity, such as myocardium or neurons, to promote healing after myocardial infarction or stroke, respectively. Despite improved ability to purify and expand stem cells, success has, thus far, been limited by difficulties in introducing and functionally integrating replacement cells at sites of damage.

Another issue arises from the immunologic reactivity of most stem cells. Although mesenchymal stem cells may be relatively immunologically privileged, most other adult stem cells, as well as ES cells (from fertilized blastocysts), express histocompatibility molecules—human leukocyte antigen (HLA) in humans (see Chapter 6)—that provoke immunologic rejection when transplanted. Hence, considerable effort has been expended to generate cells with the totipotential characteristics of ES cells from cells that can be harvested from an individual patient. This would, in principle, allow new tissues to be generated and transplanted without fear of immunologic graft rejection. To accomplish this, a handful of genes have been identified whose products can—remarkably—reprogram somatic cells into stem cells with the archetypal characteristics of ES cells. When such genes are introduced into fully differentiated cells (e.g., fibroblasts), *induced pluripotent stem (iPS) cells* are generated (Fig. 1.24). Increasingly, cells isolated from

patients carrying disease mutations are being employed to create iPS-derived cell models, which can be leveraged as “avatars” to study disease mechanisms *in vitro* at the cellular level and to screen for novel therapeutics. While not yet in practice, their differentiated progeny may be remarkable therapeutic agents (e.g., by generating insulin-secreting β cells in a patient with diabetes).

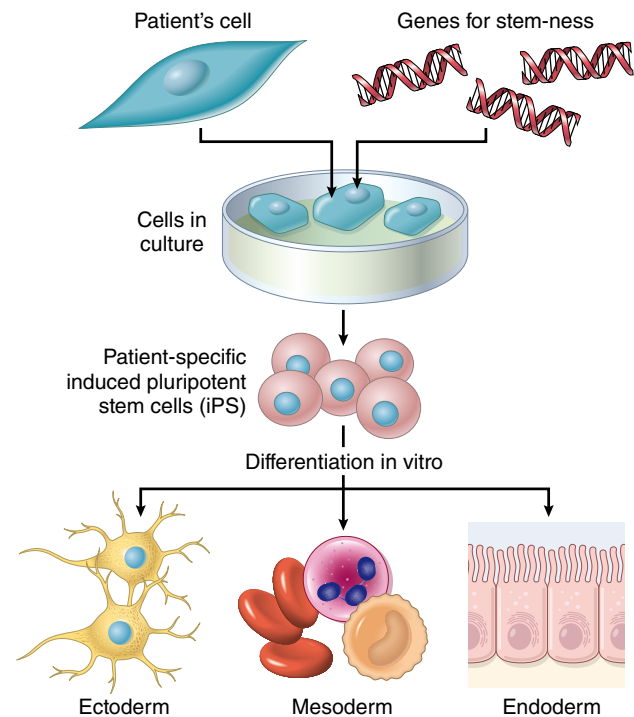


Fig. 1.24 Induced pluripotent stem cells. Genes that confer stem cell properties are introduced into a patient's differentiated cells, giving rise to stem cells that can be induced to differentiate into various lineages. (Modified from Hochedlinger K, Jaenisch R. Nuclear transplantation, embryonic stem cells, and the potential for cell therapy. *N Engl J Med.* 2003;349(3):275–286.)

CONCLUDING REMARKS

This survey of selected topics in cell biology will serve as a basis for our later discussions of pathology, and we will refer back to it throughout the book. Students should, however, remember that this summary is intentionally brief, and more information about some of the fascinating topics reviewed here can be readily found in textbooks devoted to cell and molecular biology.

ACKNOWLEDGMENT

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